tau+ (-15)
Particle 0
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-33.980, -54.272, 109.951, 127.250)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 5 simE=0.310279
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 19 dE=0.001862
raw GEN: GenParticle #16 key=2
raw SIM: n/a

Primary
Vertex 9
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.164, -0.262, 3.981, 0.218)
nIn: 1 nOut: 3
raw GEN: GenVertex #11 key=-11
raw SIM: SimVertex #74 key=9

tau- (15)
Particle 1
visibleTau
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (33.980, 54.272, -109.951, 127.250)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1343 simE=65.282730
subgraph recHits: 133 missing=1210 recoE=10.807148
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 337 dE=0.079823
raw GEN: GenParticle #17 key=3
raw SIM: n/a

Primary
Vertex 7
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.003, 0.005, 3.441, 0.198)
nIn: 1 nOut: 3
raw GEN: GenVertex #9 key=-9
raw SIM: SimVertex #72 key=7

tau+ (-15)
Particle 2
visibleTau
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (115.407, -109.393, -176.621, 237.663)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 324 simE=45.979259
subgraph recHits: 0 missing=324 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 28 dE=0.003897
raw GEN: GenParticle #18 key=4
raw SIM: n/a

Primary
Vertex 0
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.621, -0.588, 2.501, 0.240)
nIn: 1 nOut: 3
raw GEN: GenVertex #2 key=-2
raw SIM: SimVertex #66 key=1

tau- (15)
Particle 3
visibleTau
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-115.407, 109.393, 176.621, 237.663)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 673 simE=155.263596
subgraph recHits: 4 missing=669 recoE=0.137863
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 14 dE=0.001538
raw GEN: GenParticle #19 key=5
raw SIM: n/a

Primary
Vertex 1
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.193, 0.183, 3.746, 0.211)
nIn: 1 nOut: 3
raw GEN: GenVertex #3 key=-3
raw SIM: SimVertex #67 key=2

tau+ (-15)
Particle 4
visibleTau
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-51.035, -17.462, 72.607, 90.468)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 588 simE=36.719425
subgraph recHits: 39 missing=549 recoE=1.267132
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 166 dE=0.118613
raw GEN: GenParticle #20 key=6
raw SIM: n/a

Primary
Vertex 2
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.306, -0.105, 3.887, 0.216)
nIn: 1 nOut: 2
raw GEN: GenVertex #4 key=-4
raw SIM: SimVertex #68 key=3

tau- (15)
Particle 5
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (51.035, 17.462, -72.607, 90.468)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 223 simE=22.568871
subgraph recHits: 5 missing=218 recoE=0.044705
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 64 dE=0.011788
raw GEN: GenParticle #21 key=7
raw SIM: n/a

Primary
Vertex 3
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.088, 0.030, 3.326, 0.203)
nIn: 1 nOut: 3
raw GEN: GenVertex #5 key=-5
raw SIM: SimVertex #69 key=4

tau+ (-15)
Particle 6
visibleTau
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (220.400, 211.721, 1.441, 305.626)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 53 simE=1.222175
subgraph recHits: 0 missing=53 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 17 dE=0.001478
raw GEN: GenParticle #22 key=8
raw SIM: n/a

Primary
Vertex 4
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.578, 0.555, 3.454, 0.224)
nIn: 1 nOut: 2
raw GEN: GenVertex #6 key=-6
raw SIM: SimVertex #70 key=5

tau- (15)
Particle 7
visibleTau
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-220.400, -211.721, -1.441, 305.626)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 5
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 648 simE=98.882370
subgraph recHits: 0 missing=648 recoE=0.000000
direct tracker simHits: 1 dE=0.000065
subgraph tracker simHits: 23 dE=0.006939
raw GEN: GenParticle #23 key=9
raw SIM: SimTrack #520 key=1

decay
Vertex 5
domain: GEN
genEvent: 0
hasGen: yes hasSim: no
isSource: no isSink: no
x4: (-2.692, -2.586, 3.433, 0.322)
nIn: 1 nOut: 5
raw GEN: GenVertex #7 key=-7
raw SIM: n/a

tau+ (-15)
Particle 8
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (2.380, -13.871, -51.182, 53.111)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 48 simE=0.010285
subgraph recHits: 45 missing=3 recoE=2.367496
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 26 dE=0.004064
raw GEN: GenParticle #24 key=10
raw SIM: n/a

Primary
Vertex 6
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.004, -0.019, 3.380, 0.200)
nIn: 1 nOut: 3
raw GEN: GenVertex #8 key=-8
raw SIM: SimVertex #71 key=6

tau- (15)
Particle 9
visibleTau
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-2.380, 13.871, 51.182, 53.111)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 871 simE=0.346627
subgraph recHits: 733 missing=138 recoE=56.856583
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 53 dE=0.005752
raw GEN: GenParticle #25 key=11
raw SIM: n/a

Primary
Vertex 8
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.000, 0.004, 3.465, 0.198)
nIn: 1 nOut: 4
raw GEN: GenVertex #10 key=-10
raw SIM: SimVertex #73 key=8

anti-nu_tau (-16)
Particle 10
stableLegsFromUpstream
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (91.434, -87.046, -139.864, 188.412)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #26 key=12
raw SIM: n/a

pi0 (111)
Particle 11
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (22.871, -21.373, -35.092, 47.025)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 312 simE=45.677605
subgraph recHits: 0 missing=312 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 4 dE=0.001160
raw GEN: GenParticle #27 key=13
raw SIM: n/a

Primary
Vertex 10
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.622, -0.589, 2.500, 0.240)
nIn: 1 nOut: 2
raw GEN: GenVertex #12 key=-12
raw SIM: SimVertex #75 key=10

pi+ (211)
Particle 12
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (1.102, -0.974, -1.665, 2.226)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 13 simE=0.301654
direct recHits: 0 missing=13 recoE=0.000000
subgraph simHits: 13 simE=0.301654
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 24 dE=0.002737
subgraph tracker simHits: 24 dE=0.002737
checkpointId: 0
x4@checkpoint: (43.367, -115.422, -141.598, 0.000)
p4@checkpoint: (-0.200, -1.429, -1.633, 2.184)
raw GEN: GenParticle #28 key=14
raw SIM: SimTrack #521 key=3

nu_tau (16)
Particle 13
stableLegsFromUpstream
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-23.157, 21.497, 35.918, 47.838)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #29 key=15
raw SIM: n/a

pi0 (111)
Particle 14
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-61.547, 58.874, 93.693, 126.620)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 336 simE=124.380203
subgraph recHits: 3 missing=333 recoE=0.106011
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #30 key=16
raw SIM: n/a

Primary
Vertex 11
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.195, 0.185, 3.749, 0.211)
nIn: 1 nOut: 2
raw GEN: GenVertex #13 key=-13
raw SIM: SimVertex #76 key=11

pi- (-211)
Particle 15
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-30.702, 29.022, 47.010, 63.205)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 6
nCheckpoints: 1
direct simHits: 489 simE=30.860170
direct recHits: 1 missing=488 recoE=0.031853
subgraph simHits: 497 simE=30.883343
subgraph recHits: 1 missing=496 recoE=0.031853
direct tracker simHits: 14 dE=0.001538
subgraph tracker simHits: 14 dE=0.001538
checkpointId: 0
x4@checkpoint: (-90.984, 83.215, 140.714, 0.000)
p4@checkpoint: (-31.617, 27.974, 46.996, 63.173)
raw GEN: GenParticle #31 key=17
raw SIM: SimTrack #522 key=5

HadronInelastic

Vertex 341
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.967, 75.800, 140.972, 16.230)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #506 key=441

HadronInelastic

Vertex 342
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-68.021, 94.727, 170.297, 9.719)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #507 key=442

NuclearCapture
Vertex 344
domain: SIM
reason: NuclearCapture
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-97.888, 61.866, 169.886, 383.505)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #510 key=445

anti-nu_tau (-16)
Particle 16
stableLegsFromUpstream
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-16.919, -5.631, 25.387, 31.023)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #32 key=18
raw SIM: n/a

pi+ (211)
Particle 17
stableLegsFromUpstream
stableDecayProducts
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-34.117, -11.831, 47.221, 59.445)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 12
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 588 simE=36.719425
subgraph recHits: 39 missing=549 recoE=1.267132
direct tracker simHits: 2 dE=0.000146
subgraph tracker simHits: 166 dE=0.118613
raw GEN: GenParticle #33 key=19
raw SIM: SimTrack #523 key=7

HadronInelastic

Vertex 293
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-21.556, -7.392, 33.263, 1.449)
nIn: 1 nOut: 12
raw GEN: n/a
raw SIM: SimVertex #444 key=379

nu_tau (16)
Particle 18
stableLegsFromUpstream
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (16.174, 6.390, -23.582, 29.301)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #34 key=20
raw SIM: n/a

e- (11)
Particle 19
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: e- (11)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (13.537, 4.428, -19.142, 23.859)
nProdVtx: 1 nDecayVtx: 9
nParents: 1 nChildren: 9
nCheckpoints: 1
direct simHits: 5 simE=0.000674
direct recHits: 4 missing=1 recoE=0.039502
subgraph simHits: 223 simE=22.568871
subgraph recHits: 5 missing=218 recoE=0.044705
direct tracker simHits: 26 dE=0.003021
subgraph tracker simHits: 64 dE=0.011788
checkpointId: 0
x4@checkpoint: (34.492, 40.858, -293.500, 0.000)
p4@checkpoint: (-0.081, -0.057, -0.146, 0.177)
raw GEN: GenParticle #35 key=21
raw SIM: SimTrack #524 key=9

Bremsstrahlung
Vertex 247
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (9.835, 3.262, -10.475, 0.777)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #380 key=315

Bremsstrahlung
Vertex 248
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (9.848, 3.267, -10.495, 0.777)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #381 key=316

Bremsstrahlung
Vertex 249
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (9.865, 3.273, -10.518, 0.778)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #382 key=317

Bremsstrahlung
Vertex 250
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (14.151, 4.779, -16.633, 1.033)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #383 key=318

Bremsstrahlung
Vertex 251
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (22.425, 8.097, -28.643, 1.532)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #384 key=319

Bremsstrahlung
Vertex 252
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (35.691, 14.873, -48.736, 2.366)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #385 key=320

Bremsstrahlung
Vertex 253
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (47.023, 22.380, -67.136, 3.129)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #386 key=321

Bremsstrahlung
Vertex 254
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (26.489, 33.399, -132.031, 5.849)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #388 key=323

Bremsstrahlung
Vertex 255
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (48.836, 34.971, -184.601, 8.017)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #390 key=325

anti-nu_e (-12)
Particle 20
stableLegsFromUpstream
stableDecayProducts
pid: anti-nu_e (-12)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (21.324, 6.645, -29.884, 37.308)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #36 key=22
raw SIM: n/a

anti-nu_tau (-16)
Particle 21
stableLegsFromUpstream
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (152.844, 146.177, 1.661, 211.499)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #37 key=23
raw SIM: n/a

pi+ (211)
Particle 22
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (67.556, 65.544, -0.220, 94.127)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 53 simE=1.222175
direct recHits: 0 missing=53 recoE=0.000000
subgraph simHits: 53 simE=1.222175
subgraph recHits: 0 missing=53 recoE=0.000000
direct tracker simHits: 17 dE=0.001478
subgraph tracker simHits: 17 dE=0.001478
checkpointId: 0
x4@checkpoint: (89.129, 85.199, 3.181, 0.000)
p4@checkpoint: (68.501, 64.513, -0.205, 94.098)
raw GEN: GenParticle #38 key=24
raw SIM: SimTrack #525 key=12

nu_tau (16)
Particle 23
stableLegsFromUpstream
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-68.844, -65.857, -0.609, 95.273)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #39 key=25
raw SIM: n/a

pi0 (111)
Particle 24
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-39.801, -38.311, -0.665, 55.248)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 134 simE=54.157623
subgraph recHits: 0 missing=134 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #40 key=26
raw SIM: SimTrack #1159 key=303288

decay
Vertex 12
domain: GEN
genEvent: 0
hasGen: yes hasSim: no
isSource: no isSink: no
x4: (-2.692, -2.586, 3.433, 0.322)
nIn: 1 nOut: 2
raw GEN: GenVertex #14 key=-14
raw SIM: n/a

pi- (-211)
Particle 25
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-17.715, -17.062, -0.255, 24.598)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 199 simE=2.692048
direct recHits: 0 missing=199 recoE=0.000000
subgraph simHits: 201 simE=2.694066
subgraph recHits: 0 missing=201 recoE=0.000000
direct tracker simHits: 18 dE=0.002036
subgraph tracker simHits: 18 dE=0.002036
checkpointId: 0
x4@checkpoint: (-86.406, -87.959, 2.182, 0.000)
p4@checkpoint: (-16.678, -17.990, -0.242, 24.533)
raw GEN: GenParticle #41 key=27
raw SIM: SimTrack #1158 key=303287

Bremsstrahlung
Vertex 345
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-7.837, -12.106, 6.946, 2.327)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #513 key=448

Bremsstrahlung
Vertex 346
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-9.900, -13.028, 26.510, 4.543)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #514 key=449

pi- (-211)
Particle 26
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-39.880, -38.401, -0.035, 55.363)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 37 simE=0.734055
direct recHits: 0 missing=37 recoE=0.000000
subgraph simHits: 37 simE=0.734055
subgraph recHits: 0 missing=37 recoE=0.000000
direct tracker simHits: 15 dE=0.003379
subgraph tracker simHits: 15 dE=0.003379
checkpointId: 0
x4@checkpoint: (-87.780, -86.588, 3.365, 0.000)
p4@checkpoint: (-38.887, -39.362, -0.020, 55.332)
raw GEN: GenParticle #42 key=28
raw SIM: SimTrack #1157 key=303286

pi+ (211)
Particle 27
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-54.159, -52.090, 0.124, 75.144)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 7
nCheckpoints: 1
direct simHits: 478 simE=41.279270
direct recHits: 0 missing=478 recoE=0.000000
subgraph simHits: 492 simE=41.296680
subgraph recHits: 0 missing=492 recoE=0.000000
direct tracker simHits: 15 dE=0.001459
subgraph tracker simHits: 15 dE=0.001459
checkpointId: 0
x4@checkpoint: (-89.611, -84.692, 3.639, 0.000)
p4@checkpoint: (-55.065, -51.089, 0.138, 75.115)
raw GEN: GenParticle #43 key=29
raw SIM: SimTrack #1156 key=303285

HadronInelastic

Vertex 347
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-111.083, -30.614, 13.214, 12.642)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #515 key=450

HadronInelastic

Vertex 348
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-18.206, -40.770, 262.710, 85.085)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #516 key=451

HadronInelastic

Vertex 349
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-98.431, 59.984, -2.826, 21.688)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #517 key=452

HadronInelastic

Vertex 350
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-102.600, 55.352, -5.603, 24.574)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #518 key=453

HadronInelastic

Vertex 351
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (66.124, 7.666, -45.247, 22.556)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #519 key=454

anti-nu_tau (-16)
Particle 28
stableLegsFromUpstream
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.844, -2.172, -9.085, 9.379)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #44 key=30
raw SIM: n/a

mu+ (-13)
Particle 29
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: mu+ (-13)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (1.536, -9.711, -33.852, 35.251)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 48 simE=0.010285
direct recHits: 45 missing=3 recoE=2.367496
subgraph simHits: 48 simE=0.010285
subgraph recHits: 45 missing=3 recoE=2.367496
direct tracker simHits: 26 dE=0.004064
subgraph tracker simHits: 26 dE=0.004064
checkpointId: 0
x4@checkpoint: (9.266, -85.814, -293.500, 0.000)
p4@checkpoint: (0.590, -9.807, -33.800, 35.199)
raw GEN: GenParticle #45 key=31
raw SIM: SimTrack #526 key=14

Ionisation

Vertex 246
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (11.049, -164.098, -566.208, 19.978)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #379 key=314

nu_mu (14)
Particle 30
stableLegsFromUpstream
stableDecayProducts
pid: nu_mu (14)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.001, -1.987, -8.245, 8.481)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #46 key=32
raw SIM: n/a

nu_tau (16)
Particle 31
stableLegsFromUpstream
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (6.075, 8.684, -18.190, 21.052)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #47 key=33
raw SIM: n/a

pi0 (111)
Particle 32
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (9.712, 15.136, -31.217, 36.027)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 187 simE=35.414261
subgraph recHits: 0 missing=187 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #48 key=34
raw SIM: n/a

Primary
Vertex 13
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.004, 0.005, 3.440, 0.198)
nIn: 1 nOut: 2
raw GEN: GenVertex #15 key=-15
raw SIM: SimVertex #77 key=12

pi- (-211)
Particle 33
stableLegsFromUpstream
stableDecayProducts
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (18.194, 30.452, -60.545, 70.171)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 27
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1221 simE=29.868471
subgraph recHits: 133 missing=1088 recoE=10.807148
direct tracker simHits: 7 dE=0.004805
subgraph tracker simHits: 337 dE=0.079823
raw GEN: GenParticle #49 key=35
raw SIM: SimTrack #527 key=17

HadronInelastic

Vertex 79
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.462, 9.173, -14.775, 0.902)
nIn: 1 nOut: 27
raw GEN: n/a
raw SIM: SimVertex #154 key=89

nu_tau (16)
Particle 34
stableLegsFromUpstream
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.269, 0.461, 1.435, 1.531)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #50 key=36
raw SIM: n/a

pi- (-211)
Particle 35
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.955, 4.110, 15.264, 15.837)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 292 simE=0.067347
direct recHits: 232 missing=60 recoE=11.845950
subgraph simHits: 292 simE=0.067347
subgraph recHits: 232 missing=60 recoE=11.845950
direct tracker simHits: 20 dE=0.001807
subgraph tracker simHits: 20 dE=0.001807
checkpointId: 0
x4@checkpoint: (-26.156, 75.714, 293.500, 0.000)
p4@checkpoint: (-1.779, 3.815, 15.229, 15.801)
raw GEN: GenParticle #51 key=37
raw SIM: SimTrack #528 key=19

pi- (-211)
Particle 36
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.447, 1.614, 5.631, 5.876)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 55 simE=0.013826
direct recHits: 44 missing=11 recoE=6.507992
subgraph simHits: 55 simE=0.013826
subgraph recHits: 44 missing=11 recoE=6.507992
direct tracker simHits: 21 dE=0.002078
subgraph tracker simHits: 21 dE=0.002078
checkpointId: 0
x4@checkpoint: (-46.310, 73.282, 293.500, 0.000)
p4@checkpoint: (-1.283, 1.140, 5.581, 5.841)
raw GEN: GenParticle #52 key=38
raw SIM: SimTrack #529 key=20

pi+ (211)
Particle 37
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.710, 7.686, 28.852, 29.867)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 524 simE=0.265454
direct recHits: 457 missing=67 recoE=38.502655
subgraph simHits: 524 simE=0.265454
subgraph recHits: 457 missing=67 recoE=38.502655
direct tracker simHits: 20 dE=0.001866
subgraph tracker simHits: 20 dE=0.001866
checkpointId: 0
x4@checkpoint: (-2.618, 77.550, 293.500, 0.000)
p4@checkpoint: (0.167, 7.714, 28.824, 29.839)
raw GEN: GenParticle #53 key=39
raw SIM: SimTrack #530 key=21

anti-nu_tau (-16)
Particle 38
stableLegsFromUpstream
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-16.086, -24.500, 49.953, 57.916)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #54 key=40
raw SIM: n/a

mu+ (-13)
Particle 39
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: mu+ (-13)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-17.410, -29.203, 58.643, 67.786)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 5 simE=0.310279
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.310279
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 19 dE=0.001862
subgraph tracker simHits: 19 dE=0.001862
checkpointId: 0
x4@checkpoint: (-65.307, -104.585, 216.089, 0.000)
p4@checkpoint: (-18.585, -28.456, 58.586, 67.731)
raw GEN: GenParticle #55 key=41
raw SIM: SimTrack #531 key=23

Ionisation

Vertex 78
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-304.340, -443.170, 927.249, 35.854)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #153 key=88

nu_mu (14)
Particle 40
stableLegsFromUpstream
stableDecayProducts
pid: nu_mu (14)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.484, -0.569, 1.355, 1.547)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #56 key=42
raw SIM: n/a

gamma (22)
Particle 41
stableLegsFromUpstream
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (12.513, -11.785, -19.285, 25.834)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 228 simE=24.868652
subgraph recHits: 0 missing=228 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 4 dE=0.001160
raw GEN: GenParticle #57 key=43
raw SIM: SimTrack #532 key=25

PairConversion

Vertex 37
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (51.275, -48.294, -75.565, 3.729)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #105 key=40

gamma (22)
Particle 42
stableLegsFromUpstream
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (10.357, -9.589, -15.807, 21.192)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 171 simE=20.808983
subgraph recHits: 0 missing=171 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #58 key=44
raw SIM: SimTrack #533 key=26

PairConversion

Vertex 31
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.149, -77.915, -124.977, 5.941)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #99 key=34

gamma (22)
Particle 43
stableLegsFromUpstream
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-57.193, 54.697, 87.106, 117.687)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 296 simE=115.620964
subgraph recHits: 0 missing=296 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #59 key=45
raw SIM: SimTrack #534 key=27

PairConversion

Vertex 19
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.726, 80.071, 130.968, 5.945)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #83 key=18

gamma (22)
Particle 44
stableLegsFromUpstream
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-4.354, 4.177, 6.588, 8.933)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 110 simE=8.759256
subgraph recHits: 3 missing=107 recoE=0.106011
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.979, 85.355, 138.075, 0.000)
p4@checkpoint: (-4.354, 4.177, 6.588, 8.933)
raw GEN: GenParticle #60 key=46
raw SIM: SimTrack #535 key=28

PairConversion

Vertex 18
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.586, 86.896, 140.505, 6.397)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #82 key=17

gamma (22)
Particle 45
stableLegsFromUpstream
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-12.619, -12.192, -0.158, 17.548)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 82 simE=17.523069
direct recHits: 0 missing=82 recoE=0.000000
subgraph simHits: 82 simE=17.523069
subgraph recHits: 0 missing=82 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.668, -85.679, 2.359, 0.000)
p4@checkpoint: (-12.619, -12.196, -0.158, 17.550)
raw GEN: GenParticle #61 key=47
raw SIM: SimTrack #1161 key=303290

gamma (22)
Particle 46
stableLegsFromUpstream
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-27.182, -26.119, -0.507, 37.700)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 112 simE=36.634560
direct recHits: 0 missing=112 recoE=0.000000
subgraph simHits: 112 simE=36.634560
subgraph recHits: 0 missing=112 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.897, -85.442, 1.824, 0.000)
p4@checkpoint: (-27.178, -26.122, -0.507, 37.699)
raw GEN: GenParticle #62 key=48
raw SIM: SimTrack #1160 key=303289

gamma (22)
Particle 47
stableLegsFromUpstream
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (3.740, 5.725, -11.938, 13.758)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 130 simE=13.438879
subgraph recHits: 0 missing=130 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (67.437, 103.224, -211.776, 0.000)
p4@checkpoint: (3.740, 5.725, -11.938, 13.758)
raw GEN: GenParticle #63 key=49
raw SIM: SimTrack #536 key=29

PairConversion

Vertex 15
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (68.591, 104.989, -215.457, 8.613)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #79 key=14

gamma (22)
Particle 48
stableLegsFromUpstream
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (5.971, 9.411, -19.279, 22.269)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 121 simE=21.975389
subgraph recHits: 0 missing=121 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (66.059, 104.111, -209.823, 0.000)
p4@checkpoint: (5.971, 9.411, -19.279, 22.269)
raw GEN: GenParticle #64 key=50
raw SIM: SimTrack #537 key=30

PairConversion

Vertex 14
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (70.370, 110.904, -223.741, 8.951)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #78 key=13

e- (11)
Particle 49
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (2.680, 4.225, -8.655, 9.997)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 85 simE=9.882955
direct recHits: 0 missing=85 recoE=0.000000
subgraph simHits: 85 simE=9.882955
subgraph recHits: 0 missing=85 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (70.404, 110.959, -223.851, 0.000)
p4@checkpoint: (2.515, 3.964, -8.124, 9.383)
raw GEN: n/a
raw SIM: SimTrack #538 key=31

e+ (-11)
Particle 50
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (3.291, 5.186, -10.624, 12.272)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 85 simE=12.092438
direct recHits: 0 missing=85 recoE=0.000000
subgraph simHits: 85 simE=12.092438
subgraph recHits: 0 missing=85 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (70.392, 110.939, -223.812, 0.000)
p4@checkpoint: (2.033, 3.201, -6.563, 7.580)
raw GEN: n/a
raw SIM: SimTrack #539 key=32

e- (11)
Particle 51
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.269, 1.942, -4.051, 4.668)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 69 simE=4.556927
direct recHits: 0 missing=69 recoE=0.000000
subgraph simHits: 69 simE=4.557848
subgraph recHits: 0 missing=69 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (69.126, 105.813, -217.171, 0.000)
p4@checkpoint: (1.262, 1.944, -4.044, 4.661)
raw GEN: n/a
raw SIM: SimTrack #540 key=14406

Bremsstrahlung
Vertex 17
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (52.533, 101.812, -254.403, 10.385)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #81 key=16

e+ (-11)
Particle 52
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (2.471, 3.783, -7.887, 9.090)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 104 simE=8.880110
direct recHits: 0 missing=104 recoE=0.000000
subgraph simHits: 104 simE=8.881029
subgraph recHits: 0 missing=104 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (69.362, 106.166, -217.912, 0.000)
p4@checkpoint: (2.483, 3.777, -7.882, 9.086)
raw GEN: n/a
raw SIM: SimTrack #541 key=14407

Bremsstrahlung
Vertex 16
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (70.662, 92.020, -214.189, 10.147)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #80 key=15

gamma (22)
Particle 53
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000919
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000919
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (70.379, 101.240, -217.218, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #542 key=18527

gamma (22)
Particle 54
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000921
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000921
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (55.360, 110.173, -250.390, 0.000)
p4@checkpoint: (0.000, 0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #543 key=21882

e- (11)
Particle 55
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.976, 0.935, 1.476, 2.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 51 simE=1.983134
direct recHits: 0 missing=51 recoE=0.000000
subgraph simHits: 51 simE=1.983134
subgraph recHits: 0 missing=51 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-90.619, 86.928, 140.555, 0.000)
p4@checkpoint: (-0.957, 0.916, 1.447, 1.961)
raw GEN: n/a
raw SIM: SimTrack #544 key=23273

e+ (-11)
Particle 56
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-3.378, 3.242, 5.112, 6.932)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 98 simE=6.776119
direct recHits: 3 missing=95 recoE=0.106011
subgraph simHits: 98 simE=6.776119
subgraph recHits: 3 missing=95 recoE=0.106011
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-90.829, 87.130, 140.874, 0.000)
p4@checkpoint: (-1.746, 1.677, 2.642, 3.583)
raw GEN: n/a
raw SIM: SimTrack #545 key=23274

e- (11)
Particle 57
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-1.025, 0.982, 1.564, 2.112)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 42 simE=2.075948
direct recHits: 0 missing=42 recoE=0.000000
subgraph simHits: 43 simE=2.076193
subgraph recHits: 0 missing=43 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.282, 85.039, 139.204, 0.000)
p4@checkpoint: (-1.079, 0.909, 1.557, 2.101)
raw GEN: n/a
raw SIM: SimTrack #546 key=29128

Bremsstrahlung
Vertex 30
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.820, 80.160, 131.111, 5.951)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #98 key=33

e+ (-11)
Particle 58
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-56.167, 53.716, 85.542, 115.575)
nProdVtx: 1 nDecayVtx: 6
nParents: 1 nChildren: 6
nCheckpoints: 1
direct simHits: 191 simE=55.609535
direct recHits: 0 missing=191 recoE=0.000000
subgraph simHits: 290 simE=113.544792
subgraph recHits: 0 missing=290 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.105, 85.224, 139.166, 0.000)
p4@checkpoint: (-27.511, 26.409, 41.968, 56.706)
raw GEN: n/a
raw SIM: SimTrack #547 key=29129

Bremsstrahlung
Vertex 20
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.731, 80.076, 130.976, 5.945)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #84 key=19

Bremsstrahlung
Vertex 21
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.076, 80.406, 131.501, 5.969)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #85 key=20

Bremsstrahlung
Vertex 22
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.211, 80.536, 131.708, 5.978)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #86 key=21

Bremsstrahlung
Vertex 23
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-60.474, 98.791, 150.808, 8.406)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #88 key=23

Bremsstrahlung
Vertex 24
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-60.142, 98.627, 151.372, 8.368)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #89 key=24

Bremsstrahlung
Vertex 25
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-86.866, 76.458, 174.853, 7.886)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #90 key=25

gamma (22)
Particle 59
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-28.353, 27.118, 43.178, 58.340)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 5
nCheckpoints: 1
direct simHits: 187 simE=57.430328
direct recHits: 0 missing=187 recoE=0.000000
subgraph simHits: 192 simE=57.431709
subgraph recHits: 0 missing=192 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.109, 85.220, 139.166, 0.000)
p4@checkpoint: (-28.353, 27.118, 43.178, 58.340)
raw GEN: n/a
raw SIM: SimTrack #548 key=29130

Bremsstrahlung
Vertex 26
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-103.301, 54.019, 164.415, 8.218)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #92 key=27

Annihilation

Vertex 27
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (50.387, 107.070, 98.410, 12.108)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #93 key=28

Bremsstrahlung
Vertex 28
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-103.800, 51.017, 144.071, 8.495)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #95 key=30

Annihilation

Vertex 29
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-81.585, 86.910, 167.944, 7.605)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #97 key=32

gamma (22)
Particle 60
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.009, 0.009, 0.014, 0.019)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.018166
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.018166
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.109, 85.220, 139.166, 0.000)
p4@checkpoint: (-0.009, 0.009, 0.014, 0.019)
raw GEN: n/a
raw SIM: SimTrack #549 key=29132

gamma (22)
Particle 61
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.243, 0.232, 0.370, 0.499)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 21 simE=0.484230
direct recHits: 0 missing=21 recoE=0.000000
subgraph simHits: 21 simE=0.484230
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.109, 85.220, 139.166, 0.000)
p4@checkpoint: (-0.243, 0.232, 0.370, 0.499)
raw GEN: n/a
raw SIM: SimTrack #550 key=29133

gamma (22)
Particle 62
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000587
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000587
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-62.035, 106.558, 149.810, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #552 key=32184

gamma (22)
Particle 63
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000194
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000194
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-67.758, 103.013, 136.380, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #553 key=32195

gamma (22)
Particle 64
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000405
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000405
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-92.124, 81.952, 170.438, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #554 key=38723

gamma (22)
Particle 65
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000051
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000051
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.542, 76.697, 169.517, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #556 key=68380

gamma (22)
Particle 66
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000159
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000159
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (109.973, -55.758, 51.668, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #557 key=70212

gamma (22)
Particle 67
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000836
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000836
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-113.066, 49.184, 150.630, 0.000)
p4@checkpoint: (-0.001, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #560 key=71521

gamma (22)
Particle 68
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000086
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000086
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.008, 86.357, 170.785, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #562 key=82792

gamma (22)
Particle 69
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000250
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000250
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (78.538, 95.051, 117.676, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #563 key=82793

gamma (22)
Particle 70
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000245
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000245
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-74.374, 98.344, 121.642, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #564 key=104255

e- (11)
Particle 71
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (7.720, -7.147, -11.783, 15.796)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 5
nCheckpoints: 1
direct simHits: 140 simE=15.195867
direct recHits: 0 missing=140 recoE=0.000000
subgraph simHits: 145 simE=15.534009
subgraph recHits: 0 missing=145 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (90.500, -83.742, -134.630, 0.000)
p4@checkpoint: (7.615, -6.922, -11.524, 15.450)
raw GEN: n/a
raw SIM: SimTrack #565 key=105594

Bremsstrahlung
Vertex 33
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (86.401, -79.994, -128.410, 6.094)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #101 key=36

Annihilation

Vertex 34
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (98.646, -60.261, -164.341, 8.169)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #102 key=37

Bremsstrahlung
Vertex 35
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-74.011, -89.240, -128.828, 12.645)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #103 key=38

Annihilation

Vertex 36
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-74.205, -89.063, -128.657, 12.667)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #104 key=39

e+ (-11)
Particle 72
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (2.638, -2.442, -4.025, 5.396)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 77 simE=5.126217
direct recHits: 0 missing=77 recoE=0.000000
subgraph simHits: 77 simE=5.274967
subgraph recHits: 0 missing=77 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (90.391, -83.859, -134.610, 0.000)
p4@checkpoint: (2.499, -2.441, -3.902, 5.237)
raw GEN: n/a
raw SIM: SimTrack #566 key=105595

Bremsstrahlung
Vertex 32
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.166, -77.931, -125.003, 5.942)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #100 key=35

gamma (22)
Particle 73
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.074, -0.069, -0.113, 0.152)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 13 simE=0.148750
direct recHits: 0 missing=13 recoE=0.000000
subgraph simHits: 13 simE=0.148750
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (90.469, -83.775, -134.619, 0.000)
p4@checkpoint: (0.074, -0.069, -0.113, 0.152)
raw GEN: n/a
raw SIM: SimTrack #567 key=105596

gamma (22)
Particle 74
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.166, -0.153, -0.253, 0.339)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 16 simE=0.336727
direct recHits: 0 missing=16 recoE=0.000000
subgraph simHits: 16 simE=0.336727
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (90.489, -83.753, -134.630, 0.000)
p4@checkpoint: (0.166, -0.153, -0.253, 0.339)
raw GEN: n/a
raw SIM: SimTrack #568 key=109098

gamma (22)
Particle 75
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000437
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000437
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (106.435, -62.246, -170.992, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #569 key=111818

gamma (22)
Particle 76
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000376
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000376
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-78.746, -94.879, -126.672, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #570 key=116580

gamma (22)
Particle 77
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000198
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000198
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-81.754, -92.299, -135.207, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #571 key=116582

gamma (22)
Particle 78
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000405
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000405
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (115.599, -42.892, -58.933, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #572 key=116583

e- (11)
Particle 79
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (5.575, -5.250, -8.591, 11.509)
nProdVtx: 1 nDecayVtx: 8
nParents: 1 nChildren: 8
nCheckpoints: 1
direct simHits: 112 simE=10.406789
direct recHits: 0 missing=112 recoE=0.000000
subgraph simHits: 117 simE=11.193985
subgraph recHits: 0 missing=117 recoE=0.000000
direct tracker simHits: 4 dE=0.000581
subgraph tracker simHits: 4 dE=0.000581
checkpointId: 0
x4@checkpoint: (91.191, -82.988, -134.888, 0.000)
p4@checkpoint: (5.548, -4.435, -7.955, 10.664)
raw GEN: n/a
raw SIM: SimTrack #573 key=119458

Bremsstrahlung
Vertex 70
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (51.863, -48.847, -76.471, 3.769)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #143 key=78

Bremsstrahlung
Vertex 71
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (81.256, -74.878, -120.520, 5.737)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #144 key=79

Bremsstrahlung
Vertex 72
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (81.383, -74.983, -120.705, 5.746)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #145 key=80

Bremsstrahlung
Vertex 73
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (82.319, -75.763, -122.071, 5.807)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #146 key=81

Bremsstrahlung
Vertex 74
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.390, -76.651, -123.629, 5.876)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #147 key=82

Bremsstrahlung
Vertex 75
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.485, -76.729, -123.766, 5.883)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #148 key=83

Bremsstrahlung
Vertex 76
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.384, -77.470, -125.071, 5.941)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #149 key=84

Bremsstrahlung
Vertex 77
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (86.803, -79.449, -128.574, 6.098)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #150 key=85

e+ (-11)
Particle 80
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (6.938, -6.534, -10.694, 14.325)
nProdVtx: 1 nDecayVtx: 16
nParents: 1 nChildren: 16
nCheckpoints: 1
direct simHits: 53 simE=2.804753
direct recHits: 0 missing=53 recoE=0.000000
subgraph simHits: 165 simE=13.674654
subgraph recHits: 0 missing=165 recoE=0.000000
direct tracker simHits: 4 dE=0.000579
subgraph tracker simHits: 4 dE=0.000579
checkpointId: 0
x4@checkpoint: (88.480, -85.873, -134.932, 0.000)
p4@checkpoint: (1.252, -1.446, -2.147, 2.875)
raw GEN: n/a
raw SIM: SimTrack #574 key=119459

Bremsstrahlung
Vertex 38
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (79.692, -76.361, -120.402, 5.732)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #106 key=41

Bremsstrahlung
Vertex 39
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (79.702, -76.372, -120.418, 5.732)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #107 key=42

Bremsstrahlung
Vertex 40
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (79.731, -76.402, -120.464, 5.734)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #108 key=43

Bremsstrahlung
Vertex 41
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (79.780, -76.453, -120.544, 5.738)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #109 key=44

Bremsstrahlung
Vertex 42
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (81.659, -78.409, -123.591, 5.874)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #110 key=45

Bremsstrahlung
Vertex 43
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (81.666, -78.417, -123.602, 5.875)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #111 key=46

Bremsstrahlung
Vertex 44
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.369, -80.213, -126.379, 5.999)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #112 key=47

Bremsstrahlung
Vertex 45
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.376, -80.220, -126.390, 5.999)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #113 key=48

Bremsstrahlung
Vertex 46
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.415, -80.262, -126.455, 6.002)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #114 key=49

Bremsstrahlung
Vertex 47
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.514, -80.366, -126.616, 6.009)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #115 key=50

Bremsstrahlung
Vertex 48
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.575, -80.430, -126.716, 6.014)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #116 key=51

Bremsstrahlung
Vertex 49
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.767, -80.635, -127.032, 6.028)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #117 key=52

Bremsstrahlung
Vertex 50
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.919, -80.796, -127.281, 6.039)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #118 key=53

Bremsstrahlung
Vertex 51
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.152, -81.045, -127.663, 6.056)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #119 key=54

Bremsstrahlung
Vertex 52
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.828, -81.776, -128.778, 6.106)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #120 key=55

Annihilation

Vertex 53
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.654, -80.642, -126.600, 6.036)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #121 key=56

gamma (22)
Particle 81
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.007, -0.007, -0.012, 0.015)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.015419
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.015419
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.680, -85.667, -134.922, 0.000)
p4@checkpoint: (0.007, -0.007, -0.012, 0.015)
raw GEN: n/a
raw SIM: SimTrack #575 key=119463

gamma (22)
Particle 82
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.017, -0.018, -0.027, 0.037)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 1 simE=0.000491
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 4 simE=0.009358
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (84.020, -90.242, -137.536, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #576 key=119464

Bremsstrahlung
Vertex 68
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (85.319, -79.220, -137.726, 6.469)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #141 key=76

Bremsstrahlung
Vertex 69
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (85.393, -79.297, -137.793, 6.473)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #142 key=77

gamma (22)
Particle 83
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (2.138, -2.213, -3.454, 4.626)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 67 simE=4.515492
direct recHits: 0 missing=67 recoE=0.000000
subgraph simHits: 67 simE=4.515492
subgraph recHits: 0 missing=67 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.680, -85.667, -134.921, 0.000)
p4@checkpoint: (2.138, -2.213, -3.454, 4.626)
raw GEN: n/a
raw SIM: SimTrack #577 key=119465

gamma (22)
Particle 84
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.089, -1.128, -1.762, 2.359)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 51 simE=2.304676
direct recHits: 0 missing=51 recoE=0.000000
subgraph simHits: 52 simE=2.304891
subgraph recHits: 0 missing=52 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.679, -85.667, -134.938, 0.000)
p4@checkpoint: (1.089, -1.128, -1.762, 2.359)
raw GEN: n/a
raw SIM: SimTrack #578 key=119466

Annihilation

Vertex 67
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (50.418, -106.351, -168.137, 9.555)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #139 key=74

gamma (22)
Particle 85
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.022, -0.023, -0.035, 0.047)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.043002
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.043002
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.633, -85.715, -134.938, 0.000)
p4@checkpoint: (0.022, -0.023, -0.035, 0.047)
raw GEN: n/a
raw SIM: SimTrack #579 key=119467

gamma (22)
Particle 86
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.125, -0.131, -0.204, 0.272)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 30 simE=0.088986
subgraph recHits: 0 missing=30 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #580 key=119468

PairConversion

Vertex 54
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.355, -80.189, -126.356, 5.997)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #124 key=59

gamma (22)
Particle 87
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000805
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000805
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.596, -85.753, -134.916, 0.000)
p4@checkpoint: (0.000, -0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #581 key=119469

gamma (22)
Particle 88
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.004, -0.004, -0.006, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.007918
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.007918
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.604, -85.745, -134.924, 0.000)
p4@checkpoint: (0.004, -0.004, -0.006, 0.008)
raw GEN: n/a
raw SIM: SimTrack #582 key=119470

gamma (22)
Particle 89
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.027, -0.028, -0.043, 0.058)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.004587
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.004587
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.601, -85.748, -134.936, 0.000)
p4@checkpoint: (0.027, -0.028, -0.043, 0.058)
raw GEN: n/a
raw SIM: SimTrack #583 key=119472

gamma (22)
Particle 90
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.108, -0.115, -0.177, 0.237)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 18 simE=0.232569
direct recHits: 0 missing=18 recoE=0.000000
subgraph simHits: 18 simE=0.232569
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.606, -85.743, -134.946, 0.000)
p4@checkpoint: (0.108, -0.115, -0.177, 0.237)
raw GEN: n/a
raw SIM: SimTrack #584 key=119473

gamma (22)
Particle 91
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.248, -0.263, -0.407, 0.544)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 34 simE=0.537333
direct recHits: 0 missing=34 recoE=0.000000
subgraph simHits: 34 simE=0.537333
subgraph recHits: 0 missing=34 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.599, -85.750, -134.944, 0.000)
p4@checkpoint: (0.248, -0.263, -0.407, 0.544)
raw GEN: n/a
raw SIM: SimTrack #585 key=119474

gamma (22)
Particle 92
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.002, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000348
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000348
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.594, -85.756, -134.941, 0.000)
p4@checkpoint: (0.002, -0.002, -0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #586 key=119476

gamma (22)
Particle 93
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.452, -1.540, -2.381, 3.186)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 68 simE=3.106971
direct recHits: 0 missing=68 recoE=0.000000
subgraph simHits: 68 simE=3.106971
subgraph recHits: 0 missing=68 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.594, -85.756, -134.948, 0.000)
p4@checkpoint: (1.452, -1.540, -2.381, 3.186)
raw GEN: n/a
raw SIM: SimTrack #587 key=119477

gamma (22)
Particle 94
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.005, -0.005, -0.008, 0.011)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000778
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000778
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.566, -85.785, -134.919, 0.000)
p4@checkpoint: (0.005, -0.005, -0.008, 0.011)
raw GEN: n/a
raw SIM: SimTrack #588 key=119478

gamma (22)
Particle 95
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000952
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000952
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.536, -85.815, -134.912, 0.000)
p4@checkpoint: (0.000, -0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #589 key=119479

gamma (22)
Particle 96
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000492
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000492
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (74.697, -98.098, -145.271, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #590 key=124053

e- (11)
Particle 97
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.024, -0.028, -0.042, 0.056)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 4 simE=0.037638
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #593 key=124066

Bremsstrahlung
Vertex 66
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.508, -80.328, -126.567, 6.007)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #138 key=73

e+ (-11)
Particle 98
pid: e+ (-11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.099, -0.104, -0.162, 0.217)
nProdVtx: 1 nDecayVtx: 11
nParents: 1 nChildren: 13
nCheckpoints: 1
direct simHits: 6 simE=0.002922
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 26 simE=0.051348
subgraph recHits: 0 missing=26 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (85.564, -88.778, -134.477, 0.000)
p4@checkpoint: (-0.015, -0.142, -0.125, 0.189)
raw GEN: n/a
raw SIM: SimTrack #594 key=124067

Bremsstrahlung
Vertex 55
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.467, -80.313, -126.549, 6.006)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #125 key=60

Bremsstrahlung
Vertex 56
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.745, -80.725, -127.059, 6.030)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #127 key=62

Bremsstrahlung
Vertex 57
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (71.799, -90.881, -165.169, 7.920)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #129 key=64

Bremsstrahlung
Vertex 58
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (72.730, -91.111, -165.877, 7.960)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #130 key=65

Bremsstrahlung
Vertex 59
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (74.176, -92.178, -166.540, 8.025)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #131 key=66

Annihilation

Vertex 60
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (74.134, -93.353, -192.981, 9.021)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #132 key=67

Bremsstrahlung
Vertex 61
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (70.359, -92.116, -174.739, 8.314)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #133 key=68

Bremsstrahlung
Vertex 62
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (70.373, -92.170, -174.791, 8.317)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #134 key=69

Bremsstrahlung
Vertex 63
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (70.408, -92.312, -174.866, 8.322)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #135 key=70

Annihilation

Vertex 64
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (70.200, -93.272, -175.110, 8.357)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #136 key=71

Annihilation

Vertex 65
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (70.357, -91.988, -174.568, 8.307)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #137 key=72

gamma (22)
Particle 99
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, -0.003, -0.005, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000132
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000132
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.286, -86.072, -135.469, 0.000)
p4@checkpoint: (0.003, -0.003, -0.005, 0.006)
raw GEN: n/a
raw SIM: SimTrack #595 key=124072

gamma (22)
Particle 100
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000331
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000331
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (87.472, -86.900, -133.642, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #597 key=124075

gamma (22)
Particle 101
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001169
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001169
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (80.969, -92.989, -170.560, 0.000)
p4@checkpoint: (0.001, -0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #599 key=124109

gamma (22)
Particle 102
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000110
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000110
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (80.161, -93.686, -171.948, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #600 key=124110

gamma (22)
Particle 103
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.011, -0.016, -0.006, 0.020)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.020201
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.020201
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (77.088, -96.230, -168.003, 0.000)
p4@checkpoint: (0.011, -0.016, -0.006, 0.020)
raw GEN: n/a
raw SIM: SimTrack #601 key=124111

gamma (22)
Particle 104
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000116
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000116
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.852, -82.256, -205.199, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #602 key=124113

gamma (22)
Particle 105
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000211
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000211
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (33.560, -118.645, -179.871, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #603 key=124114

gamma (22)
Particle 106
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.003, -0.002, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003192
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003192
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (72.542, -99.702, -182.252, 0.000)
p4@checkpoint: (0.001, -0.003, -0.002, 0.004)
raw GEN: n/a
raw SIM: SimTrack #604 key=124141

gamma (22)
Particle 107
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001102
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001102
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (73.728, -98.829, -180.963, 0.000)
p4@checkpoint: (0.000, -0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #605 key=124142

gamma (22)
Particle 108
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000285
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000285
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (70.450, -101.191, -176.009, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #606 key=124144

gamma (22)
Particle 109
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.007, -0.018, -0.008, 0.021)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.021029
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.021029
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (65.696, -104.340, -180.217, 0.000)
p4@checkpoint: (-0.007, -0.018, -0.008, 0.021)
raw GEN: n/a
raw SIM: SimTrack #608 key=124147

gamma (22)
Particle 110
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000076
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000076
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (86.899, -87.473, -169.062, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #609 key=124170

gamma (22)
Particle 111
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000473
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000473
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (54.246, -110.726, -177.674, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #610 key=124171

gamma (22)
Particle 112
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.021, -0.020, -0.032, 0.044)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.037638
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.037638
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.947, -85.389, -134.791, 0.000)
p4@checkpoint: (0.021, -0.020, -0.032, 0.044)
raw GEN: n/a
raw SIM: SimTrack #611 key=124187

gamma (22)
Particle 113
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000215
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000215
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (44.375, -115.038, -161.255, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #612 key=125376

gamma (22)
Particle 114
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000786
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000786
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (87.468, -86.904, -157.785, 0.000)
p4@checkpoint: (0.000, -0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #614 key=128697

gamma (22)
Particle 115
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.005, -0.006, -0.002, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.008081
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.008081
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (89.759, -84.536, -139.130, 0.000)
p4@checkpoint: (0.005, -0.006, -0.002, 0.008)
raw GEN: n/a
raw SIM: SimTrack #615 key=128699

gamma (22)
Particle 116
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.019, -0.018, -0.029, 0.039)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.039101
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.039101
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (89.812, -84.479, -134.880, 0.000)
p4@checkpoint: (0.019, -0.018, -0.029, 0.039)
raw GEN: n/a
raw SIM: SimTrack #616 key=128718

gamma (22)
Particle 117
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.091, -0.076, -0.133, 0.178)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=0.176890
direct recHits: 0 missing=9 recoE=0.000000
subgraph simHits: 9 simE=0.176890
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.097, -83.091, -134.891, 0.000)
p4@checkpoint: (0.091, -0.076, -0.133, 0.178)
raw GEN: n/a
raw SIM: SimTrack #617 key=128720

gamma (22)
Particle 118
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.068, -0.057, -0.099, 0.133)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=0.129665
direct recHits: 0 missing=9 recoE=0.000000
subgraph simHits: 9 simE=0.129665
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.098, -83.090, -134.888, 0.000)
p4@checkpoint: (0.068, -0.057, -0.099, 0.133)
raw GEN: n/a
raw SIM: SimTrack #618 key=128721

gamma (22)
Particle 119
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.161, -0.133, -0.234, 0.314)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 21 simE=0.281020
direct recHits: 0 missing=21 recoE=0.000000
subgraph simHits: 21 simE=0.281020
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.115, -83.072, -134.886, 0.000)
p4@checkpoint: (0.161, -0.133, -0.234, 0.314)
raw GEN: n/a
raw SIM: SimTrack #619 key=128722

gamma (22)
Particle 120
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.019, -0.016, -0.028, 0.037)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.037335
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.037335
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.132, -83.054, -134.875, 0.000)
p4@checkpoint: (0.019, -0.016, -0.028, 0.037)
raw GEN: n/a
raw SIM: SimTrack #620 key=128723

gamma (22)
Particle 121
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.052, -0.043, -0.076, 0.102)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 16 simE=0.100071
direct recHits: 0 missing=16 recoE=0.000000
subgraph simHits: 16 simE=0.100071
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.135, -83.050, -134.881, 0.000)
p4@checkpoint: (0.052, -0.043, -0.076, 0.102)
raw GEN: n/a
raw SIM: SimTrack #621 key=128724

gamma (22)
Particle 122
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001185
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001185
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.149, -83.034, -134.881, 0.000)
p4@checkpoint: (0.001, -0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #622 key=128725

gamma (22)
Particle 123
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.012, -0.009, -0.017, 0.023)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.021927
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.021927
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.176, -83.005, -134.888, 0.000)
p4@checkpoint: (0.012, -0.009, -0.017, 0.023)
raw GEN: n/a
raw SIM: SimTrack #623 key=128727

e- (11)
Particle 124
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.000, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #626 key=137766

pdg(331)
Particle 125
pid: pdg(331)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.237, 0.727, -0.987, 1.573)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 11
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 87 simE=0.996936
subgraph recHits: 0 missing=87 recoE=0.000000
direct tracker simHits: 6 dE=0.000619
subgraph tracker simHits: 26 dE=0.003786
raw GEN: n/a
raw SIM: SimTrack #627 key=200639

Decay
Vertex 216
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.462, 9.173, -14.775, 0.902)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #345 key=280

HadronAtRest

Vertex 241
domain: SIM
reason: HadronAtRest
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.668, 9.060, -14.939, 0.931)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #374 key=309

HadronInelastic

Vertex 242
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (96.227, 54.826, 68.617, 37.733)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #375 key=310

HadronInelastic

Vertex 243
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.861, -35.103, 0.505, 12.765)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #376 key=311

HadronInelastic

Vertex 244
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-7.198, 36.539, -43.306, 8.023)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #377 key=312

pi- (-211)
Particle 126
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.018, 0.222, -0.297, 0.396)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 19 dE=0.003064
subgraph tracker simHits: 19 dE=0.003064
checkpointId: 0
x4@checkpoint: (10.632, 11.349, -293.500, 0.000)
p4@checkpoint: (-0.063, 0.233, -0.200, 0.343)
raw GEN: n/a
raw SIM: SimTrack #628 key=200640

pi+ (211)
Particle 127
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.115, 0.182, -0.078, 0.268)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 11 dE=0.006434
subgraph tracker simHits: 11 dE=0.006434
raw GEN: n/a
raw SIM: SimTrack #629 key=200641

n (2112)
Particle 128
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.598, 1.426, -5.656, 6.120)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 47 simE=0.012606
direct recHits: 12 missing=35 recoE=3.984685
subgraph simHits: 47 simE=0.012606
subgraph recHits: 12 missing=35 recoE=3.984685
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (82.331, 80.430, -293.500, 0.000)
p4@checkpoint: (1.549, 1.454, -5.662, 6.120)
raw GEN: n/a
raw SIM: SimTrack #630 key=200642

pi+ (211)
Particle 129
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.590, 3.830, -6.225, 7.481)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 35 simE=0.545938
direct recHits: 5 missing=30 recoE=0.732953
subgraph simHits: 35 simE=0.545938
subgraph recHits: 5 missing=30 recoE=0.732953
direct tracker simHits: 13 dE=0.001897
subgraph tracker simHits: 13 dE=0.001897
checkpointId: 0
x4@checkpoint: (64.075, 105.344, -184.293, 0.000)
p4@checkpoint: (2.673, 3.137, -6.175, 7.426)
raw GEN: n/a
raw SIM: SimTrack #631 key=200643

pi0 (111)
Particle 130
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.379, 2.199, -4.070, 4.829)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 60 simE=4.289836
subgraph recHits: 6 missing=54 recoE=0.210858
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 24 dE=0.003260
raw GEN: n/a
raw SIM: SimTrack #632 key=200644

Decay
Vertex 205
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.462, 9.173, -14.775, 0.902)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #329 key=264

pi+ (211)
Particle 131
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.133, 0.359, -0.578, 0.707)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000069
direct recHits: 2 missing=0 recoE=0.113850
subgraph simHits: 2 simE=0.000069
subgraph recHits: 2 missing=0 recoE=0.113850
direct tracker simHits: 27 dE=0.003038
subgraph tracker simHits: 27 dE=0.003038
checkpointId: 0
x4@checkpoint: (10.715, -15.481, -293.500, 0.000)
p4@checkpoint: (-0.168, 0.302, -0.568, 0.679)
raw GEN: n/a
raw SIM: SimTrack #633 key=200645

pi+ (211)
Particle 132
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.071, 0.285, -0.435, 0.544)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 17 simE=0.001751
direct recHits: 16 missing=1 recoE=0.399733
subgraph simHits: 17 simE=0.001751
subgraph recHits: 16 missing=1 recoE=0.399733
direct tracker simHits: 14 dE=0.001808
subgraph tracker simHits: 14 dE=0.001808
checkpointId: 0
x4@checkpoint: (45.499, 13.145, -293.500, 0.000)
p4@checkpoint: (0.261, -0.242, -0.324, 0.501)
raw GEN: n/a
raw SIM: SimTrack #634 key=200646

pi0 (111)
Particle 133
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.224, 1.426, -3.505, 3.979)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 70 simE=3.601525
subgraph recHits: 9 missing=61 recoE=0.130594
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #635 key=200647

Decay
Vertex 204
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.462, 9.173, -14.775, 0.902)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #328 key=263

pi- (-211)
Particle 134
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.268, 2.613, -5.123, 5.891)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 82 simE=2.951101
direct recHits: 0 missing=82 recoE=0.000000
subgraph simHits: 82 simE=2.951101
subgraph recHits: 0 missing=82 recoE=0.000000
direct tracker simHits: 17 dE=0.001980
subgraph tracker simHits: 17 dE=0.001980
checkpointId: 0
x4@checkpoint: (31.055, 119.325, -216.338, 0.000)
p4@checkpoint: (0.024, 2.858, -5.085, 5.835)
raw GEN: n/a
raw SIM: SimTrack #636 key=200648

pi0 (111)
Particle 135
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (2.538, 4.784, -8.822, 10.353)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 186 simE=9.514767
subgraph recHits: 5 missing=181 recoE=0.116921
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 40 dE=0.004948
raw GEN: n/a
raw SIM: SimTrack #637 key=200649

Decay
Vertex 151
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.462, 9.173, -14.775, 0.902)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #246 key=181

pi- (-211)
Particle 136
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.406, 0.556, -0.773, 1.045)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 28 simE=0.135924
direct recHits: 7 missing=21 recoE=1.480641
subgraph simHits: 28 simE=0.135924
subgraph recHits: 7 missing=21 recoE=1.480641
direct tracker simHits: 14 dE=0.001549
subgraph tracker simHits: 14 dE=0.001549
checkpointId: 0
x4@checkpoint: (-92.973, 42.847, -293.500, 0.000)
p4@checkpoint: (0.153, -0.508, -0.731, 0.914)
raw GEN: n/a
raw SIM: SimTrack #638 key=200650

pi0 (111)
Particle 137
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.028, 0.518, -0.585, 0.793)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 25 simE=0.195438
subgraph recHits: 11 missing=14 recoE=0.404817
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 34 dE=0.005434
raw GEN: n/a
raw SIM: SimTrack #639 key=200651

Decay
Vertex 144
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.462, 9.173, -14.775, 0.902)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #232 key=167

pi+ (211)
Particle 138
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.297, 2.201, -4.128, 4.857)
nProdVtx: 1 nDecayVtx: 9
nParents: 1 nChildren: 47
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 197 simE=1.293242
subgraph recHits: 1 missing=196 recoE=0.020566
direct tracker simHits: 31 dE=0.007179
subgraph tracker simHits: 37 dE=0.014048
raw GEN: n/a
raw SIM: SimTrack #640 key=200652

Ionisation

Vertex 124
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.094, 58.817, -117.589, 4.939)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #208 key=143

HadronInelastic

Vertex 125
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (78.575, 85.030, -186.025, 7.629)
nIn: 1 nOut: 30
raw GEN: n/a
raw SIM: SimVertex #209 key=144

HadronInelastic

Vertex 126
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (20.105, -11.908, -160.343, 47.826)
nIn: 1 nOut: 4
raw GEN: n/a
raw SIM: SimVertex #210 key=145

HadronInelastic

Vertex 127
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.811, 82.757, -190.454, 9.063)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #211 key=146

HadronInelastic

Vertex 128
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.869, 81.139, -184.516, 8.683)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #213 key=148

HadronInelastic

Vertex 129
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (40.786, 108.227, -190.787, 13.813)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #214 key=149

HadronInelastic

Vertex 130
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (38.783, 110.015, -185.853, 18.472)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #215 key=150

HadronInelastic

Vertex 131
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (78.451, 84.792, -184.389, 8.115)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #216 key=151

HadronInelastic

Vertex 132
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (68.917, 95.494, -172.771, 10.096)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #217 key=152

pi0 (111)
Particle 139
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.260, 0.435, -0.539, 0.752)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 25 simE=0.686939
subgraph recHits: 0 missing=25 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #641 key=200653

Decay
Vertex 123
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.462, 9.173, -14.775, 0.902)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #207 key=142

pi- (-211)
Particle 140
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.239, 1.521, -4.270, 4.702)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 47 simE=0.078638
direct recHits: 15 missing=32 recoE=1.898880
subgraph simHits: 48 simE=0.079257
subgraph recHits: 15 missing=33 recoE=1.898880
direct tracker simHits: 21 dE=0.002371
subgraph tracker simHits: 21 dE=0.002371
checkpointId: 0
x4@checkpoint: (46.558, 114.172, -264.754, 0.000)
p4@checkpoint: (0.102, 1.945, -4.217, 4.647)
raw GEN: n/a
raw SIM: SimTrack #642 key=200654

Bremsstrahlung
Vertex 122
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (29.353, -111.900, -220.618, 22.162)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #206 key=141

pi0 (111)
Particle 141
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.256, -0.182, -0.366, 0.501)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 25 simE=0.226062
subgraph recHits: 0 missing=25 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 2 dE=0.000145
raw GEN: n/a
raw SIM: SimTrack #643 key=200655

Decay
Vertex 102
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.462, 9.173, -14.775, 0.902)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #182 key=117

gamma (22)
Particle 142
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.290, 0.659, -0.605, 0.941)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 32 simE=0.824393
subgraph recHits: 0 missing=32 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #644 key=200656

PairConversion

Vertex 86
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (47.636, 105.160, -102.910, 5.471)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #162 key=97

pi- (-211)
Particle 143
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.535, 1.221, -2.256, 2.624)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 62 simE=1.180943
direct recHits: 0 missing=62 recoE=0.000000
subgraph simHits: 64 simE=1.184229
subgraph recHits: 0 missing=64 recoE=0.000000
direct tracker simHits: 19 dE=0.002168
subgraph tracker simHits: 19 dE=0.002168
checkpointId: 0
x4@checkpoint: (-6.880, 123.108, -215.104, 0.000)
p4@checkpoint: (-0.787, 1.097, -2.167, 2.557)
raw GEN: n/a
raw SIM: SimTrack #645 key=200657

HadronInelastic

Vertex 85
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-18.714, 18.515, -180.851, 23.204)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #161 key=96

pi+ (211)
Particle 144
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.774, 1.881, -3.478, 4.336)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 119 simE=1.984674
direct recHits: 0 missing=119 recoE=0.000000
subgraph simHits: 119 simE=1.984733
subgraph recHits: 0 missing=119 recoE=0.000000
direct tracker simHits: 14 dE=0.002319
subgraph tracker simHits: 14 dE=0.002319
checkpointId: 0
x4@checkpoint: (101.481, 70.033, -168.925, 0.000)
p4@checkpoint: (2.427, 0.802, -3.420, 4.272)
raw GEN: n/a
raw SIM: SimTrack #646 key=200658

Bremsstrahlung
Vertex 84
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (96.057, 68.145, -161.260, 6.998)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #160 key=95

pi- (-211)
Particle 145
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.050, 0.030, -0.405, 0.433)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 19 dE=0.008231
subgraph tracker simHits: 19 dE=0.008231
checkpointId: 0
x4@checkpoint: (-7.683, 1.374, -293.500, 0.000)
p4@checkpoint: (0.028, -0.079, -0.343, 0.379)
raw GEN: n/a
raw SIM: SimTrack #647 key=200659

p (2212)
Particle 146
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.344, 0.818, -1.695, 2.131)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 30 simE=0.012313
subgraph recHits: 12 missing=18 recoE=0.572329
direct tracker simHits: 8 dE=0.000747
subgraph tracker simHits: 22 dE=0.002200
raw GEN: n/a
raw SIM: SimTrack #648 key=200660

HadronInelastic

Vertex 83
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (21.745, 33.374, -71.596, 3.281)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #158 key=93

pi- (-211)
Particle 147
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.637, 1.803, -2.850, 3.435)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 65 simE=0.989388
direct recHits: 0 missing=65 recoE=0.000000
subgraph simHits: 65 simE=0.989388
subgraph recHits: 0 missing=65 recoE=0.000000
direct tracker simHits: 13 dE=0.001470
subgraph tracker simHits: 13 dE=0.001470
checkpointId: 0
x4@checkpoint: (4.620, 123.213, -187.930, 0.000)
p4@checkpoint: (-0.648, 1.777, -2.801, 3.382)
raw GEN: n/a
raw SIM: SimTrack #649 key=200661

pi+ (211)
Particle 148
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.029, 1.259, -2.384, 2.889)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 67 simE=0.324067
subgraph recHits: 32 missing=35 recoE=0.740322
direct tracker simHits: 13 dE=0.001454
subgraph tracker simHits: 21 dE=0.004867
raw GEN: n/a
raw SIM: SimTrack #650 key=200662

HadronInelastic

Vertex 80
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (87.341, 58.298, -157.924, 6.684)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #155 key=90

n (2112)
Particle 149
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.244, -0.080, -0.033, 0.975)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000120
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000120
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-119.095, -31.927, -31.695, 0.000)
p4@checkpoint: (-0.244, -0.080, -0.033, 0.975)
raw GEN: n/a
raw SIM: SimTrack #651 key=200664

n (2112)
Particle 150
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.089, 0.080, 0.082, 0.951)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000009
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000009
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-84.655, 89.646, 68.413, 0.000)
p4@checkpoint: (-0.089, 0.080, 0.082, 0.951)
raw GEN: n/a
raw SIM: SimTrack #652 key=200667

n (2112)
Particle 151
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.225, -0.059, -0.022, 0.968)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.017851
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.017851
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.450, -21.276, -26.067, 0.000)
p4@checkpoint: (0.225, -0.059, -0.022, 0.968)
raw GEN: n/a
raw SIM: SimTrack #653 key=200668

pi+ (211)
Particle 152
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.349, -0.308, -2.044, 2.101)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 37 simE=0.004972
direct recHits: 32 missing=5 recoE=0.740322
subgraph simHits: 37 simE=0.004972
subgraph recHits: 32 missing=5 recoE=0.740322
direct tracker simHits: 6 dE=0.000515
subgraph tracker simHits: 6 dE=0.000515
checkpointId: 0
x4@checkpoint: (100.835, 31.987, -293.500, 0.000)
p4@checkpoint: (0.041, -0.443, -2.045, 2.098)
raw GEN: n/a
raw SIM: SimTrack #654 key=200889

n (2112)
Particle 153
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.230, 0.572, -0.376, 1.692)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 29 simE=0.317688
subgraph recHits: 0 missing=29 recoE=0.000000
direct tracker simHits: 2 dE=0.002898
subgraph tracker simHits: 2 dE=0.002898
raw GEN: n/a
raw SIM: SimTrack #655 key=200890

HadronInelastic

Vertex 81
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (87.364, 58.308, -157.931, 6.686)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #156 key=91

gamma (22)
Particle 154
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, -0.000, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001406
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001406
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-121.151, 22.919, -279.435, 0.000)
p4@checkpoint: (-0.002, -0.000, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #656 key=200891

n (2112)
Particle 155
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.930, 0.563, -0.131, 1.443)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 24 simE=0.316440
subgraph recHits: 0 missing=24 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #657 key=200894

HadronInelastic

Vertex 82
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (93.888, 62.254, -158.848, 7.023)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #157 key=92

n (2112)
Particle 156
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.030, 0.050, -0.007, 0.941)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 11 simE=0.000672
direct recHits: 0 missing=11 recoE=0.000000
subgraph simHits: 11 simE=0.000672
subgraph recHits: 0 missing=11 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (97.711, 75.203, -159.891, 0.000)
p4@checkpoint: (0.031, 0.050, -0.004, 0.941)
raw GEN: n/a
raw SIM: SimTrack #658 key=200897

gamma (22)
Particle 157
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000576
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000576
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (84.003, 90.257, -131.949, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #659 key=200899

n (2112)
Particle 158
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.900, 0.609, -0.057, 1.438)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 23 simE=0.315903
direct recHits: 0 missing=23 recoE=0.000000
subgraph simHits: 23 simE=0.315903
subgraph recHits: 0 missing=23 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (102.705, 68.224, -159.409, 0.000)
p4@checkpoint: (0.900, 0.609, -0.057, 1.438)
raw GEN: n/a
raw SIM: SimTrack #660 key=201027

gamma (22)
Particle 159
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.004, 0.002, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000537
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000537
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.119, 17.024, -133.501, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #661 key=201028

p (2212)
Particle 160
caloBoundary
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.663, 0.386, -1.720, 2.104)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 28 simE=0.006858
direct recHits: 12 missing=16 recoE=0.572329
subgraph simHits: 28 simE=0.006858
subgraph recHits: 12 missing=16 recoE=0.572329
direct tracker simHits: 14 dE=0.001453
subgraph tracker simHits: 14 dE=0.001453
checkpointId: 0
x4@checkpoint: (110.065, 15.002, -293.500, 0.000)
p4@checkpoint: (0.415, -0.588, -1.716, 2.084)
raw GEN: n/a
raw SIM: SimTrack #662 key=203618

gamma (22)
Particle 161
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, 0.004, -0.001, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.005455
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.005455
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (77.533, 95.873, -90.398, 0.000)
p4@checkpoint: (0.003, 0.004, -0.001, 0.005)
raw GEN: n/a
raw SIM: SimTrack #663 key=203619

gamma (22)
Particle 162
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000059
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000059
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (103.261, 67.379, -174.136, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #666 key=204652

n (2112)
Particle 163
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.105, -0.161, -0.057, 0.961)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002729
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002729
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-87.446, -86.925, -218.129, 0.000)
p4@checkpoint: (-0.105, -0.161, -0.057, 0.961)
raw GEN: n/a
raw SIM: SimTrack #667 key=204884

gamma (22)
Particle 164
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000556
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000556
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (74.435, -98.297, -278.433, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #668 key=204885

e- (11)
Particle 165
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.267, 0.607, -0.557, 0.866)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 21 simE=0.813490
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #669 key=204938

Bremsstrahlung
Vertex 99
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (47.676, 105.253, -102.995, 5.475)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #176 key=111

Bremsstrahlung
Vertex 100
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (47.678, 105.257, -102.999, 5.476)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #177 key=112

Bremsstrahlung
Vertex 101
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (46.968, 106.636, -105.106, 5.566)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #178 key=113

e+ (-11)
Particle 166
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.023, 0.052, -0.048, 0.075)
nProdVtx: 1 nDecayVtx: 11
nParents: 1 nChildren: 11
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 13 simE=0.010903
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #670 key=204939

Bremsstrahlung
Vertex 87
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (47.650, 105.193, -102.943, 5.473)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #163 key=98

Bremsstrahlung
Vertex 88
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (50.364, 104.077, -104.243, 5.650)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #164 key=99

Bremsstrahlung
Vertex 89
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (50.461, 104.132, -104.274, 5.654)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #165 key=100

Bremsstrahlung
Vertex 90
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (50.446, 104.187, -104.234, 5.656)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #166 key=101

Annihilation

Vertex 91
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (50.392, 104.169, -104.191, 5.658)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #167 key=102

Bremsstrahlung
Vertex 92
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (47.759, 105.473, -103.197, 5.486)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #168 key=103

Annihilation

Vertex 93
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (48.241, 106.114, -103.231, 5.525)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #169 key=104

Bremsstrahlung
Vertex 94
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (47.718, 105.356, -103.120, 5.481)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #170 key=105

Bremsstrahlung
Vertex 95
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (47.728, 105.451, -103.256, 5.486)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #171 key=106

Bremsstrahlung
Vertex 96
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (46.274, 106.050, -106.422, 5.623)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #172 key=107

Bremsstrahlung
Vertex 97
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (46.790, 106.497, -105.589, 5.584)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #174 key=109

gamma (22)
Particle 167
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.001, 0.003, -0.003, 0.005)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 1 simE=0.000267
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 2 simE=0.001470
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (43.217, 115.478, -99.841, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #671 key=204940

Bremsstrahlung
Vertex 98
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (48.861, 108.124, -105.931, 5.618)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #175 key=110

gamma (22)
Particle 168
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001347
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001347
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (66.127, 104.068, -113.732, 0.000)
p4@checkpoint: (0.001, -0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #672 key=204950

gamma (22)
Particle 169
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000514
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000514
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (51.443, 112.056, -101.503, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #673 key=204952

gamma (22)
Particle 170
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000393
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000393
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (46.990, 113.995, -96.404, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #674 key=204953

gamma (22)
Particle 171
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.000, -0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000356
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000356
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (51.360, 112.094, -106.308, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #675 key=204957

gamma (22)
Particle 172
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000435
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000435
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.346, 112.553, -108.838, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #676 key=204970

gamma (22)
Particle 173
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000511
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000511
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (49.826, 112.784, -102.072, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #677 key=204975

gamma (22)
Particle 174
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000538
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000538
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (49.897, 112.753, -112.022, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #678 key=204978

gamma (22)
Particle 175
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.002, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.003389
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.003389
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (49.350, 112.993, -112.467, 0.000)
p4@checkpoint: (0.000, 0.002, -0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #679 key=204979

gamma (22)
Particle 176
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, -0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001640
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001640
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (84.805, -89.504, -230.150, 0.000)
p4@checkpoint: (0.000, -0.001, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #680 key=204991

gamma (22)
Particle 177
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000312
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000312
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (60.122, 107.649, -100.223, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #682 key=205003

gamma (22)
Particle 178
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001202
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001202
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.672, 113.287, -111.637, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #683 key=205006

gamma (22)
Particle 179
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.250, 0.573, -0.531, 0.821)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 20 simE=0.813065
direct recHits: 0 missing=20 recoE=0.000000
subgraph simHits: 20 simE=0.813065
subgraph recHits: 0 missing=20 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.772, 112.361, -109.586, 0.000)
p4@checkpoint: (0.250, 0.573, -0.531, 0.821)
raw GEN: n/a
raw SIM: SimTrack #684 key=205007

gamma (22)
Particle 180
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000316
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000316
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.587, 112.445, -108.882, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #685 key=205008

gamma (22)
Particle 181
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000109
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000109
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (49.237, 113.042, -116.155, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #686 key=205015

gamma (22)
Particle 182
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.241, -0.171, -0.262, 0.395)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 25 simE=0.226062
subgraph recHits: 0 missing=25 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #690 key=205306

PairConversion

Vertex 103
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (100.008, -57.990, -117.540, 6.071)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #184 key=119

gamma (22)
Particle 183
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.015, -0.010, -0.104, 0.105)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 2 dE=0.000145
subgraph tracker simHits: 2 dE=0.000145
checkpointId: 0
x4@checkpoint: (46.187, -17.756, -293.500, 0.000)
p4@checkpoint: (0.016, -0.007, -0.090, 0.092)
raw GEN: n/a
raw SIM: SimTrack #691 key=205307

e- (11)
Particle 184
pid: e- (11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.092, -0.066, -0.100, 0.150)
nProdVtx: 1 nDecayVtx: 9
nParents: 1 nChildren: 9
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 10 simE=0.026810
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (108.895, -57.834, -127.091, 0.000)
p4@checkpoint: (0.066, 0.053, -0.085, 0.120)
raw GEN: n/a
raw SIM: SimTrack #693 key=205435

Bremsstrahlung
Vertex 113
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (100.757, -58.488, -118.383, 6.112)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #195 key=130

Bremsstrahlung
Vertex 114
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (106.668, -45.455, -143.201, 7.270)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #198 key=133

Bremsstrahlung
Vertex 115
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (106.615, -45.442, -143.252, 7.272)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #199 key=134

Bremsstrahlung
Vertex 116
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (103.817, -48.452, -148.686, 7.537)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #200 key=135

Bremsstrahlung
Vertex 117
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (104.926, -48.851, -150.282, 7.604)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #201 key=136

Bremsstrahlung
Vertex 118
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (105.030, -48.744, -150.462, 7.612)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #202 key=137

Bremsstrahlung
Vertex 119
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (105.054, -48.610, -150.544, 7.617)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #203 key=138

Bremsstrahlung
Vertex 120
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (104.184, -44.761, -147.527, 7.550)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #204 key=139

Annihilation

Vertex 121
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (104.172, -44.306, -147.091, 7.582)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #205 key=140

e+ (-11)
Particle 185
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.150, -0.106, -0.162, 0.245)
nProdVtx: 1 nDecayVtx: 9
nParents: 1 nChildren: 9
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 16 simE=0.199252
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #694 key=205436

Bremsstrahlung
Vertex 104
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (100.051, -58.021, -117.588, 6.073)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #185 key=120

Bremsstrahlung
Vertex 105
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (100.096, -58.052, -117.636, 6.076)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #186 key=121

Bremsstrahlung
Vertex 106
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (100.106, -58.059, -117.647, 6.076)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #187 key=122

Bremsstrahlung
Vertex 107
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (100.115, -58.066, -117.657, 6.077)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #188 key=123

Bremsstrahlung
Vertex 108
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (100.134, -58.080, -117.677, 6.078)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #189 key=124

Bremsstrahlung
Vertex 109
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (101.571, -59.538, -119.304, 6.165)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #190 key=125

Bremsstrahlung
Vertex 110
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (99.066, -60.041, -133.453, 7.104)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #192 key=127

Bremsstrahlung
Vertex 111
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (97.779, -62.404, -135.147, 7.265)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #193 key=128

Annihilation

Vertex 112
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (97.912, -62.083, -134.982, 7.288)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #194 key=129

gamma (22)
Particle 186
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.001, -0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003137
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003137
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (106.330, -62.425, -124.478, 0.000)
p4@checkpoint: (0.002, -0.001, -0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #695 key=205437

gamma (22)
Particle 187
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000194
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000194
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (107.233, -60.859, -103.249, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #696 key=205438

gamma (22)
Particle 188
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001113
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001113
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (106.319, -62.444, -124.294, 0.000)
p4@checkpoint: (0.001, -0.001, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #697 key=205439

gamma (22)
Particle 189
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.089, -0.063, -0.095, 0.145)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=0.162347
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=0.162347
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (106.294, -62.485, -124.297, 0.000)
p4@checkpoint: (0.089, -0.063, -0.095, 0.145)
raw GEN: n/a
raw SIM: SimTrack #698 key=205440

gamma (22)
Particle 190
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.011, -0.008, -0.013, 0.019)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.016699
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.016699
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (106.245, -62.570, -124.513, 0.000)
p4@checkpoint: (0.011, -0.008, -0.013, 0.019)
raw GEN: n/a
raw SIM: SimTrack #699 key=205441

gamma (22)
Particle 191
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001311
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001311
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (104.709, -65.106, -123.982, 0.000)
p4@checkpoint: (0.001, -0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #700 key=205443

gamma (22)
Particle 192
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.005, -0.007, -0.007, 0.011)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.009957
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.009957
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (103.942, -66.324, -139.804, 0.000)
p4@checkpoint: (0.005, -0.007, -0.007, 0.011)
raw GEN: n/a
raw SIM: SimTrack #702 key=205466

gamma (22)
Particle 193
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, 0.003, -0.000, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.004208
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.004208
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-103.652, 66.777, -154.174, 0.000)
p4@checkpoint: (-0.004, 0.003, -0.000, 0.005)
raw GEN: n/a
raw SIM: SimTrack #703 key=205469

gamma (22)
Particle 194
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000286
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000286
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (104.838, -64.900, -145.646, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #704 key=205475

gamma (22)
Particle 195
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.011, -0.006, -0.013, 0.018)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.018992
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.018992
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (106.694, -61.801, -125.035, 0.000)
p4@checkpoint: (0.011, -0.006, -0.013, 0.018)
raw GEN: n/a
raw SIM: SimTrack #705 key=205551

gamma (22)
Particle 196
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000221
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000221
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.962, -24.304, -293.500, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #708 key=205560

gamma (22)
Particle 197
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.031, 0.009, -0.029, 0.043)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000164
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000164
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-50.561, -1.804, -293.500, 0.000)
p4@checkpoint: (-0.031, 0.009, -0.029, 0.043)
raw GEN: n/a
raw SIM: SimTrack #709 key=205561

gamma (22)
Particle 198
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.002, -0.004, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.005477
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.005477
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (109.953, -55.797, -160.677, 0.000)
p4@checkpoint: (0.002, -0.002, -0.004, 0.005)
raw GEN: n/a
raw SIM: SimTrack #710 key=205566

gamma (22)
Particle 199
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000142
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000142
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (112.939, -49.474, -164.346, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #711 key=205569

gamma (22)
Particle 200
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.002, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001039
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001039
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.924, 9.624, -169.678, 0.000)
p4@checkpoint: (0.000, 0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #712 key=205570

gamma (22)
Particle 201
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000275
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000275
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (115.039, -44.374, -157.839, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #713 key=205571

gamma (22)
Particle 202
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000145
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000145
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.072, -17.358, -27.736, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #714 key=205583

gamma (22)
Particle 203
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000355
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000355
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (109.275, -57.113, -134.787, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #715 key=205585

gamma (22)
Particle 204
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000619
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000619
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (39.658, -116.748, -217.849, 0.000)
p4@checkpoint: (0.001, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #717 key=206205

gamma (22)
Particle 205
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.129, 0.168, -0.316, 0.380)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 14 simE=0.339902
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 14 simE=0.339902
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (73.814, 98.764, -182.721, 0.000)
p4@checkpoint: (0.129, 0.168, -0.316, 0.380)
raw GEN: n/a
raw SIM: SimTrack #718 key=208168

gamma (22)
Particle 206
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.131, 0.267, -0.223, 0.372)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 11 simE=0.347037
direct recHits: 0 missing=11 recoE=0.000000
subgraph simHits: 11 simE=0.347037
subgraph recHits: 0 missing=11 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (55.201, 110.253, -99.235, 0.000)
p4@checkpoint: (0.131, 0.267, -0.223, 0.372)
raw GEN: n/a
raw SIM: SimTrack #719 key=208169

e- (11)
Particle 207
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.008, 0.004, -0.020, 0.022)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.000570
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 5 dE=0.000755
subgraph tracker simHits: 5 dE=0.000755
checkpointId: 0
x4@checkpoint: (46.890, 60.411, -293.500, 0.000)
p4@checkpoint: (0.006, 0.002, -0.009, 0.011)
raw GEN: n/a
raw SIM: SimTrack #720 key=208402

Bremsstrahlung
Vertex 143
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (44.945, 60.266, -132.032, 5.466)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #231 key=166

pi+ (211)
Particle 208
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.689, 0.468, -0.795, 1.160)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 50 simE=0.446142
direct recHits: 0 missing=50 recoE=0.000000
subgraph simHits: 50 simE=0.446142
subgraph recHits: 0 missing=50 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (85.355, 88.980, -193.515, 0.000)
p4@checkpoint: (0.739, 0.378, -0.789, 1.154)
raw GEN: n/a
raw SIM: SimTrack #721 key=208414

p (2212)
Particle 209
caloBoundary
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.268, 0.608, -0.300, 1.188)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.201660
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.201660
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (74.482, 98.262, -191.932, 0.000)
p4@checkpoint: (-0.107, 0.617, -0.258, 1.157)
raw GEN: n/a
raw SIM: SimTrack #722 key=208415

n (2112)
Particle 210
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.068, -0.393, 0.037, 1.021)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.000016
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.000016
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (43.648, -115.316, -167.223, 0.000)
p4@checkpoint: (-0.068, -0.393, 0.037, 1.021)
raw GEN: n/a
raw SIM: SimTrack #723 key=208416

n (2112)
Particle 211
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.078, -0.028, 0.426, 1.035)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.005273
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.005273
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-61.534, -106.848, 169.278, 0.000)
p4@checkpoint: (-0.146, -0.197, 0.353, 1.033)
raw GEN: n/a
raw SIM: SimTrack #724 key=208417

n (2112)
Particle 212
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.155, 0.255, 0.059, 0.988)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000033
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000033
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (67.547, 103.152, -181.812, 0.000)
p4@checkpoint: (-0.155, 0.255, 0.059, 0.988)
raw GEN: n/a
raw SIM: SimTrack #725 key=208418

n (2112)
Particle 213
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.127, 0.486, 0.187, 1.082)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 23 simE=0.042657
direct recHits: 0 missing=23 recoE=0.000000
subgraph simHits: 23 simE=0.042657
subgraph recHits: 0 missing=23 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (75.279, 97.652, -181.165, 0.000)
p4@checkpoint: (-0.127, 0.486, 0.187, 1.082)
raw GEN: n/a
raw SIM: SimTrack #726 key=208420

n (2112)
Particle 214
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.046, -0.109, -0.372, 1.017)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 4 simE=0.010317
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 1 dE=0.000002
subgraph tracker simHits: 1 dE=0.000069
raw GEN: n/a
raw SIM: SimTrack #728 key=208422

HadronInelastic

Vertex 142
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (57.865, 78.440, -224.539, 11.557)
nIn: 1 nOut: 4
raw GEN: n/a
raw SIM: SimVertex #230 key=165

n (2112)
Particle 215
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.380, 0.182, -0.226, 1.054)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2 simE=0.000148
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #729 key=208423

HadronInelastic

Vertex 141
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.758, 114.137, -222.203, 13.258)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #229 key=164

p (2212)
Particle 216
caloBoundary
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.526, 0.168, 0.162, 1.101)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.121666
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.121666
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (87.270, 87.102, -183.551, 0.000)
p4@checkpoint: (0.514, 0.073, 0.144, 1.082)
raw GEN: n/a
raw SIM: SimTrack #730 key=208424

pi0 (111)
Particle 217
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.124, 0.089, 0.076, 0.217)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 15 simE=0.130023
subgraph recHits: 0 missing=15 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 2 dE=0.002371
raw GEN: n/a
raw SIM: SimTrack #731 key=208425

Decay
Vertex 133
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (78.575, 85.030, -186.025, 7.629)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #218 key=153

p (2212)
Particle 218
caloBoundary
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.549, 0.230, 0.072, 1.113)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.130322
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.130322
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (86.774, 87.597, -185.134, 0.000)
p4@checkpoint: (0.552, 0.124, 0.057, 1.097)
raw GEN: n/a
raw SIM: SimTrack #732 key=208426

n (2112)
Particle 219
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.016, 0.028, 0.337, 0.999)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000030
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000030
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-43.422, 109.219, 293.500, 0.000)
p4@checkpoint: (-0.085, 0.016, 0.327, 0.999)
raw GEN: n/a
raw SIM: SimTrack #733 key=208428

n (2112)
Particle 220
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.418, 0.228, -0.564, 1.195)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.001417
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.001417
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (85.526, 88.815, -195.405, 0.000)
p4@checkpoint: (0.418, 0.228, -0.564, 1.195)
raw GEN: n/a
raw SIM: SimTrack #734 key=208429

n (2112)
Particle 221
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.169, 0.135, -0.300, 1.009)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.000887
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.000887
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (84.518, 89.775, -196.586, 0.000)
p4@checkpoint: (0.169, 0.135, -0.300, 1.009)
raw GEN: n/a
raw SIM: SimTrack #735 key=208430

n (2112)
Particle 222
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.608, -0.087, -0.057, 1.124)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002004
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002004
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-108.628, 58.334, -203.700, 0.000)
p4@checkpoint: (-0.608, -0.087, -0.057, 1.124)
raw GEN: n/a
raw SIM: SimTrack #736 key=208431

n (2112)
Particle 223
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.087, -0.049, 0.107, 0.951)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.002207
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.002207
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-120.527, -26.002, 55.491, 0.000)
p4@checkpoint: (-0.107, -0.062, 0.080, 0.951)
raw GEN: n/a
raw SIM: SimTrack #737 key=208432

n (2112)
Particle 224
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.107, 0.009, -0.212, 0.969)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.000512
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.000512
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 2 dE=0.003401
subgraph tracker simHits: 2 dE=0.003401
checkpointId: 0
x4@checkpoint: (77.923, 95.555, -223.949, 0.000)
p4@checkpoint: (0.031, 0.079, -0.059, 0.945)
raw GEN: n/a
raw SIM: SimTrack #738 key=208434

n (2112)
Particle 225
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.201, -0.255, -0.117, 1.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.044190
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.044190
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (113.350, 48.526, -203.404, 0.000)
p4@checkpoint: (0.273, -0.186, -0.097, 1.001)
raw GEN: n/a
raw SIM: SimTrack #740 key=208437

n (2112)
Particle 226
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.257, -0.351, 0.053, 1.037)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000063
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000063
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (120.215, 27.408, -165.923, 0.000)
p4@checkpoint: (0.223, -0.314, 0.209, 1.037)
raw GEN: n/a
raw SIM: SimTrack #741 key=208438

n (2112)
Particle 227
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.063, 0.272, -0.110, 0.986)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 22 simE=0.110855
direct recHits: 0 missing=22 recoE=0.000000
subgraph simHits: 22 simE=0.110855
subgraph recHits: 0 missing=22 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (66.151, 104.053, -187.663, 0.000)
p4@checkpoint: (-0.154, 0.195, 0.003, 0.972)
raw GEN: n/a
raw SIM: SimTrack #742 key=208440

n (2112)
Particle 228
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.122, -0.293, -0.340, 1.048)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000180
direct recHits: 1 missing=0 recoE=0.020566
subgraph simHits: 1 simE=0.000180
subgraph recHits: 1 missing=0 recoE=0.020566
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (117.067, -7.676, -293.500, 0.000)
p4@checkpoint: (0.122, -0.293, -0.340, 1.048)
raw GEN: n/a
raw SIM: SimTrack #743 key=208441

n (2112)
Particle 229
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.053, 0.089, -0.074, 0.948)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.006164
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.006164
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (82.519, 91.616, -191.509, 0.000)
p4@checkpoint: (0.053, 0.089, -0.074, 0.948)
raw GEN: n/a
raw SIM: SimTrack #744 key=208442

n (2112)
Particle 230
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.089, -0.099, 0.045, 0.950)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=0.000004
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=0.000004
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (85.490, 88.850, -187.425, 0.000)
p4@checkpoint: (0.009, 0.010, -0.002, 0.940)
raw GEN: n/a
raw SIM: SimTrack #745 key=208443

n (2112)
Particle 231
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.171, -0.044, 0.013, 0.956)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.002102
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.002102
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (92.614, 81.398, -184.949, 0.000)
p4@checkpoint: (0.171, -0.044, 0.013, 0.956)
raw GEN: n/a
raw SIM: SimTrack #746 key=208444

n (2112)
Particle 232
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.128, -0.053, 0.140, 0.960)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.006886
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.006886
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.330, 78.199, -177.542, 0.000)
p4@checkpoint: (0.177, -0.072, -0.014, 0.959)
raw GEN: n/a
raw SIM: SimTrack #747 key=208445

n (2112)
Particle 233
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.157, -0.020, 0.132, 0.962)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.000415
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.000415
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-90.441, 83.805, -124.696, 0.000)
p4@checkpoint: (-0.100, 0.046, -0.151, 0.958)
raw GEN: n/a
raw SIM: SimTrack #748 key=208446

n (2112)
Particle 234
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.011, -0.110, -0.160, 0.960)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000046
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000046
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (70.895, 11.264, -293.500, 0.000)
p4@checkpoint: (-0.011, -0.110, -0.160, 0.960)
raw GEN: n/a
raw SIM: SimTrack #752 key=208459

n (2112)
Particle 235
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.019, 0.034, 0.066, 0.943)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.008819
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.008819
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (72.457, 99.764, -170.881, 0.000)
p4@checkpoint: (-0.018, 0.071, 0.006, 0.942)
raw GEN: n/a
raw SIM: SimTrack #753 key=208460

n (2112)
Particle 236
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.035, 0.013, -0.044, 0.941)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000047
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000047
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (86.417, 87.949, -195.877, 0.000)
p4@checkpoint: (0.035, 0.013, -0.044, 0.941)
raw GEN: n/a
raw SIM: SimTrack #754 key=208468

gamma (22)
Particle 237
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.002, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002383
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002383
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (101.182, 70.464, -168.086, 0.000)
p4@checkpoint: (0.002, -0.002, 0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #755 key=208470

gamma (22)
Particle 238
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.002, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000057
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000057
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-1.807, -123.287, -64.187, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #756 key=208504

n (2112)
Particle 239
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.008, -0.003, 0.054, 0.941)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.000008
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.000008
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.427, -112.517, 18.013, 0.000)
p4@checkpoint: (0.002, -0.005, 0.003, 0.940)
raw GEN: n/a
raw SIM: SimTrack #757 key=208505

gamma (22)
Particle 240
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000499
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000499
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-37.936, -117.319, -217.922, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #758 key=208506

gamma (22)
Particle 241
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000224
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000224
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (103.011, -67.761, -196.273, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #759 key=208507

gamma (22)
Particle 242
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000972
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000972
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (70.670, 101.038, -189.087, 0.000)
p4@checkpoint: (-0.001, 0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #760 key=208518

gamma (22)
Particle 243
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000494
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000494
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (90.459, 83.785, -186.259, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #762 key=208520

n (2112)
Particle 244
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.013, 0.020, 0.013, 0.940)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (87.076, 87.296, -191.590, 0.000)
p4@checkpoint: (0.011, 0.015, -0.005, 0.940)
raw GEN: n/a
raw SIM: SimTrack #763 key=208521

gamma (22)
Particle 245
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, -0.001, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000718
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000718
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.150, 16.801, -136.708, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #766 key=208624

gamma (22)
Particle 246
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.003, 0.004, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003165
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003165
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (37.372, 117.500, -178.640, 0.000)
p4@checkpoint: (-0.000, 0.003, 0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #768 key=208639

gamma (22)
Particle 247
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.002, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.002002
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.002002
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (39.028, 116.960, -196.263, 0.000)
p4@checkpoint: (-0.000, 0.002, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #769 key=208642

gamma (22)
Particle 248
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000321
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000321
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.325, 112.562, -182.197, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #772 key=208654

gamma (22)
Particle 249
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000106
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000106
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.016, 114.789, -219.925, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #773 key=208655

gamma (22)
Particle 250
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.000, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001011
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001011
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 1 dE=0.000273
subgraph tracker simHits: 1 dE=0.000273
checkpointId: 0
x4@checkpoint: (-95.518, 77.969, -52.103, 0.000)
p4@checkpoint: (-0.001, 0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #774 key=208667

n (2112)
Particle 251
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.053, 0.145, 0.058, 0.954)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000026
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000026
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (70.882, 100.889, -170.597, 0.000)
p4@checkpoint: (0.053, 0.145, 0.058, 0.954)
raw GEN: n/a
raw SIM: SimTrack #776 key=208976

gamma (22)
Particle 252
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.005, 0.000, 0.001, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.004805
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.004805
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (77.941, 95.541, -170.224, 0.000)
p4@checkpoint: (0.005, 0.000, 0.001, 0.005)
raw GEN: n/a
raw SIM: SimTrack #777 key=208978

gamma (22)
Particle 253
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000795
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000795
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-85.661, -88.685, -232.444, 0.000)
p4@checkpoint: (-0.000, -0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #779 key=208980

gamma (22)
Particle 254
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.014, 0.084, 0.064, 0.107)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.098816
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.098816
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (76.600, 96.619, -177.194, 0.000)
p4@checkpoint: (-0.014, 0.084, 0.064, 0.107)
raw GEN: n/a
raw SIM: SimTrack #780 key=209251

gamma (22)
Particle 255
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.109, 0.004, 0.011, 0.110)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 10 simE=0.031207
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 2 dE=0.001715
subgraph tracker simHits: 2 dE=0.002371
raw GEN: n/a
raw SIM: SimTrack #781 key=209252

PairConversion

Vertex 134
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (78.038, 85.051, -185.970, 7.647)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #219 key=154

Bremsstrahlung
Vertex 135
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (75.594, 87.303, -185.396, 7.769)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #221 key=156

Annihilation

Vertex 136
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (75.547, 87.821, -185.029, 7.791)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #222 key=157

e- (11)
Particle 256
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.079, 0.003, 0.008, 0.079)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 7 simE=0.025063
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 2 dE=0.000656
subgraph tracker simHits: 2 dE=0.000656
raw GEN: n/a
raw SIM: SimTrack #782 key=209253

Bremsstrahlung
Vertex 137
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (78.033, 85.051, -185.969, 7.647)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #224 key=159

Bremsstrahlung
Vertex 138
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (75.303, 84.586, -185.693, 7.741)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #225 key=160

Bremsstrahlung
Vertex 139
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (76.483, 74.642, -184.426, 8.192)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #226 key=161

Bremsstrahlung
Vertex 140
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (76.589, 74.628, -184.414, 8.195)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #227 key=162

gamma (22)
Particle 257
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.002, 0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003068
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003068
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (73.789, 98.783, -183.471, 0.000)
p4@checkpoint: (-0.000, 0.002, 0.000, 0.002)
raw GEN: n/a
raw SIM: SimTrack #784 key=209262

gamma (22)
Particle 258
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.003, 0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.003077
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.003077
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (75.503, 97.479, -182.326, 0.000)
p4@checkpoint: (-0.000, 0.003, 0.001, 0.003)
raw GEN: n/a
raw SIM: SimTrack #786 key=209269

gamma (22)
Particle 259
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, 0.000, 0.000, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002850
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002850
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-79.033, 94.640, -169.554, 0.000)
p4@checkpoint: (-0.003, 0.000, 0.000, 0.003)
raw GEN: n/a
raw SIM: SimTrack #788 key=209289

gamma (22)
Particle 260
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.010, -0.004, 0.001, 0.011)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.008814
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.008814
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-123.298, 0.764, -166.625, 0.000)
p4@checkpoint: (-0.010, -0.004, 0.001, 0.011)
raw GEN: n/a
raw SIM: SimTrack #789 key=209292

gamma (22)
Particle 261
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, 0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001504
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001504
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (100.913, 70.849, -180.876, 0.000)
p4@checkpoint: (0.001, -0.000, 0.000, 0.002)
raw GEN: n/a
raw SIM: SimTrack #790 key=209300

gamma (22)
Particle 262
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.014, -0.002, 0.002, 0.014)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.011895
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.011895
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (100.137, 71.940, -181.630, 0.000)
p4@checkpoint: (0.014, -0.002, 0.002, 0.014)
raw GEN: n/a
raw SIM: SimTrack #791 key=209301

n (2112)
Particle 263
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.333, 0.201, -0.176, 1.032)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000148
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000148
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (3.942, 123.237, -228.373, 0.000)
p4@checkpoint: (-0.334, 0.222, -0.147, 1.032)
raw GEN: n/a
raw SIM: SimTrack #793 key=209362

n (2112)
Particle 264
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.026, -0.166, -0.049, 0.956)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.008285
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.008285
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (27.304, -120.239, -283.641, 0.000)
p4@checkpoint: (-0.026, -0.166, -0.049, 0.956)
raw GEN: n/a
raw SIM: SimTrack #794 key=209378

n (2112)
Particle 265
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.037, 0.043, -0.034, 0.942)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (65.592, 104.406, -232.730, 0.000)
p4@checkpoint: (-0.006, 0.030, 0.003, 0.940)
raw GEN: n/a
raw SIM: SimTrack #796 key=209382

gamma (22)
Particle 266
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.002, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.002031
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.002031
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-11.472, -122.765, -61.921, 0.000)
p4@checkpoint: (-0.001, -0.002, 0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #797 key=209383

gamma (22)
Particle 267
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 1 dE=0.000067
subgraph tracker simHits: 1 dE=0.000067
checkpointId: 0
x4@checkpoint: (74.665, 42.615, -293.500, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #798 key=209384

gamma (22)
Particle 268
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000570
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000570
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-32.888, 118.833, -260.305, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #799 key=210129

gamma (22)
Particle 269
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.035, 0.183, -0.125, 0.225)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=0.191424
direct recHits: 0 missing=9 recoE=0.000000
subgraph simHits: 9 simE=0.191424
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-15.869, 122.275, -92.042, 0.000)
p4@checkpoint: (-0.035, 0.183, -0.125, 0.225)
raw GEN: n/a
raw SIM: SimTrack #800 key=210145

gamma (22)
Particle 270
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.007, 0.334, -0.460, 0.569)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 16 simE=0.004014
subgraph recHits: 11 missing=5 recoE=0.404817
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 34 dE=0.005434
raw GEN: n/a
raw SIM: SimTrack #801 key=210146

PairConversion

Vertex 145
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (6.126, 41.384, -59.074, 2.729)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #233 key=168

e- (11)
Particle 271
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.005, 0.194, -0.268, 0.331)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 2 simE=0.000341
direct recHits: 1 missing=1 recoE=0.021319
subgraph simHits: 7 simE=0.002473
subgraph recHits: 5 missing=2 recoE=0.263600
direct tracker simHits: 9 dE=0.001053
subgraph tracker simHits: 9 dE=0.001053
checkpointId: 0
x4@checkpoint: (-18.313, 38.170, -293.500, 0.000)
p4@checkpoint: (-0.046, 0.003, -0.062, 0.078)
raw GEN: n/a
raw SIM: SimTrack #802 key=210147

Bremsstrahlung
Vertex 150
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-20.496, 29.885, -152.871, 6.586)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #242 key=177

e+ (-11)
Particle 272
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.003, 0.140, -0.193, 0.238)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 6 simE=0.000820
direct recHits: 6 missing=0 recoE=0.141217
subgraph simHits: 9 simE=0.001541
subgraph recHits: 6 missing=3 recoE=0.141217
direct tracker simHits: 23 dE=0.003951
subgraph tracker simHits: 25 dE=0.004381
checkpointId: 0
x4@checkpoint: (7.471, 44.488, -293.500, 0.000)
p4@checkpoint: (0.018, 0.124, -0.119, 0.173)
raw GEN: n/a
raw SIM: SimTrack #803 key=210148

Bremsstrahlung
Vertex 146
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (30.987, 40.487, -109.742, 4.861)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #234 key=169

Bremsstrahlung
Vertex 147
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (6.912, 44.976, -154.198, 6.911)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #237 key=172

gamma (22)
Particle 273
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000099
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000099
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 1 dE=0.000373
subgraph tracker simHits: 1 dE=0.000373
checkpointId: 0
x4@checkpoint: (108.451, 19.303, 293.500, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #804 key=210156

gamma (22)
Particle 274
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.004, 0.015, -0.015, 0.022)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2 simE=0.000622
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 1 dE=0.000057
subgraph tracker simHits: 1 dE=0.000057
checkpointId: 0
x4@checkpoint: (23.570, 121.026, -230.211, 0.000)
p4@checkpoint: (-0.001, 0.004, -0.005, 0.006)
raw GEN: n/a
raw SIM: SimTrack #807 key=210163

Bremsstrahlung
Vertex 148
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (25.107, 115.070, -223.337, 10.252)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #238 key=173

Bremsstrahlung
Vertex 149
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (23.663, 115.391, -222.905, 10.451)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #240 key=175

gamma (22)
Particle 275
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000044
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000044
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (22.103, 121.303, -223.234, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #808 key=210473

gamma (22)
Particle 276
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000578
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000578
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-121.888, -18.606, -222.950, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #810 key=210477

gamma (22)
Particle 277
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.105, -0.071, -0.182, 0.222)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.002132
direct recHits: 4 missing=1 recoE=0.242282
subgraph simHits: 5 simE=0.002132
subgraph recHits: 4 missing=1 recoE=0.242282
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (60.772, -25.132, -293.500, 0.000)
p4@checkpoint: (0.105, -0.071, -0.182, 0.222)
raw GEN: n/a
raw SIM: SimTrack #812 key=210496

gamma (22)
Particle 278
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.206, 2.307, -4.116, 4.870)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 141 simE=4.118099
subgraph recHits: 5 missing=136 recoE=0.116921
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 40 dE=0.004948
raw GEN: n/a
raw SIM: SimTrack #816 key=212206

PairConversion

Vertex 152
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (10.844, 19.469, -33.141, 1.627)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #247 key=182

gamma (22)
Particle 279
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.332, 2.477, -4.706, 5.483)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 53 simE=5.396664
direct recHits: 0 missing=53 recoE=0.000000
subgraph simHits: 53 simE=5.396664
subgraph recHits: 0 missing=53 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (58.810, 108.371, -203.265, 0.000)
p4@checkpoint: (1.332, 2.477, -4.706, 5.483)
raw GEN: n/a
raw SIM: SimTrack #817 key=212207

e- (11)
Particle 280
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.137, 2.178, -3.883, 4.595)
nProdVtx: 1 nDecayVtx: 25
nParents: 1 nChildren: 26
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 133 simE=4.114950
subgraph recHits: 1 missing=132 recoE=0.008778
direct tracker simHits: 16 dE=0.001821
subgraph tracker simHits: 17 dE=0.001903
raw GEN: n/a
raw SIM: SimTrack #818 key=213923

Bremsstrahlung
Vertex 157
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (13.865, 25.471, -43.754, 2.046)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #259 key=194

Bremsstrahlung
Vertex 158
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.511, 33.440, -57.588, 2.592)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #260 key=195

Bremsstrahlung
Vertex 159
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.533, 33.493, -57.678, 2.596)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #261 key=196

Bremsstrahlung
Vertex 160
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (25.196, 72.508, -120.731, 5.088)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #262 key=197

Bremsstrahlung
Vertex 161
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (24.448, 80.622, -133.503, 5.593)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #263 key=198

Bremsstrahlung
Vertex 162
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.981, 113.047, -193.277, 7.957)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #264 key=199

Bremsstrahlung
Vertex 163
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.883, 113.131, -193.478, 7.965)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #265 key=200

Bremsstrahlung
Vertex 164
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.752, 113.242, -193.746, 7.976)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #266 key=201

Bremsstrahlung
Vertex 165
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.587, 115.733, -200.044, 8.225)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #267 key=202

Bremsstrahlung
Vertex 166
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.542, 115.767, -200.133, 8.229)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #268 key=203

Bremsstrahlung
Vertex 167
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.515, 115.787, -200.186, 8.231)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #269 key=204

Bremsstrahlung
Vertex 168
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.502, 115.796, -200.210, 8.232)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #270 key=205

Bremsstrahlung
Vertex 169
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.481, 115.813, -200.253, 8.234)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #272 key=207

Bremsstrahlung
Vertex 170
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.426, 115.854, -200.360, 8.238)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #273 key=208

Bremsstrahlung
Vertex 171
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.374, 115.895, -200.464, 8.242)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #274 key=209

Bremsstrahlung
Vertex 172
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.368, 115.900, -200.477, 8.243)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #275 key=210

Bremsstrahlung
Vertex 173
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.348, 115.915, -200.517, 8.244)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #277 key=212

Bremsstrahlung
Vertex 174
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.294, 115.956, -200.623, 8.248)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #278 key=213

Bremsstrahlung
Vertex 175
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (1.386, 116.584, -202.409, 8.318)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #279 key=214

Bremsstrahlung
Vertex 176
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (1.341, 116.613, -202.497, 8.322)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #281 key=216

Bremsstrahlung
Vertex 177
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (0.911, 116.862, -203.439, 8.357)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #282 key=217

Bremsstrahlung
Vertex 178
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-0.832, 116.052, -208.572, 8.546)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #284 key=219

Annihilation

Vertex 186
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.272, 115.988, -200.700, 8.251)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #300 key=235

Bremsstrahlung
Vertex 187
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.368, 115.925, -201.396, 8.278)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #301 key=236

Annihilation

Vertex 188
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.189, 115.762, -201.653, 8.285)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #302 key=237

e+ (-11)
Particle 281
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.069, 0.129, -0.233, 0.275)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 8 simE=0.003149
subgraph recHits: 4 missing=4 recoE=0.108143
direct tracker simHits: 24 dE=0.002736
subgraph tracker simHits: 25 dE=0.003045
checkpointId: 0
x4@checkpoint: (27.457, 14.187, -293.500, 0.000)
p4@checkpoint: (-0.049, -0.048, -0.122, 0.140)
raw GEN: n/a
raw SIM: SimTrack #819 key=213924

Bremsstrahlung
Vertex 153
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (10.850, 19.480, -33.161, 1.628)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #248 key=183

Bremsstrahlung
Vertex 154
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (16.791, 25.388, -164.066, 6.815)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #252 key=187

Bremsstrahlung
Vertex 155
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (28.584, 24.108, -183.035, 7.573)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #254 key=189

Bremsstrahlung
Vertex 156
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (19.849, 24.438, -262.108, 10.708)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #257 key=192

gamma (22)
Particle 282
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001663
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001663
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (58.921, 108.310, -194.553, 0.000)
p4@checkpoint: (0.000, 0.001, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #820 key=213925

gamma (22)
Particle 283
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, -0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000649
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000649
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 1 dE=0.000309
subgraph tracker simHits: 1 dE=0.000309
checkpointId: 0
x4@checkpoint: (111.204, 53.259, -260.447, 0.000)
p4@checkpoint: (0.001, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #824 key=213940

gamma (22)
Particle 284
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.008, -0.006, -0.014, 0.017)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000205
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000205
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (86.588, -20.803, -293.500, 0.000)
p4@checkpoint: (0.008, -0.006, -0.014, 0.017)
raw GEN: n/a
raw SIM: SimTrack #826 key=213942

gamma (22)
Particle 285
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.022, 0.010, -0.039, 0.046)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.000632
direct recHits: 4 missing=0 recoE=0.108143
subgraph simHits: 4 simE=0.000632
subgraph recHits: 4 missing=0 recoE=0.108143
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (37.483, 32.253, -293.500, 0.000)
p4@checkpoint: (0.022, 0.010, -0.039, 0.046)
raw GEN: n/a
raw SIM: SimTrack #829 key=213950

gamma (22)
Particle 286
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.089, 0.184, -0.322, 0.382)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 30 simE=0.195090
subgraph recHits: 0 missing=30 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #831 key=214238

PairConversion

Vertex 191
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (52.585, 105.348, -183.975, 7.582)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #306 key=241

gamma (22)
Particle 287
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.446, 1.036, -1.780, 2.107)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 32 simE=2.129013
direct recHits: 1 missing=31 recoE=0.008778
subgraph simHits: 35 simE=2.129866
subgraph recHits: 1 missing=34 recoE=0.008778
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (51.376, 112.087, -192.755, 0.000)
p4@checkpoint: (0.446, 1.036, -1.780, 2.107)
raw GEN: n/a
raw SIM: SimTrack #832 key=214239

Annihilation

Vertex 190
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (63.302, 62.290, -224.668, 10.763)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #305 key=240

gamma (22)
Particle 288
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.020, 0.047, -0.080, 0.095)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.086933
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.086933
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (51.344, 112.101, -192.362, 0.000)
p4@checkpoint: (0.020, 0.047, -0.080, 0.095)
raw GEN: n/a
raw SIM: SimTrack #833 key=214240

gamma (22)
Particle 289
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.008, 0.452, -0.711, 0.843)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 18 simE=0.867928
direct recHits: 0 missing=18 recoE=0.000000
subgraph simHits: 18 simE=0.867928
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (24.374, 120.867, -196.763, 0.000)
p4@checkpoint: (-0.008, 0.452, -0.711, 0.843)
raw GEN: n/a
raw SIM: SimTrack #834 key=214243

gamma (22)
Particle 290
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.004, -0.006, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.008378
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.008378
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (17.414, 122.064, -199.646, 0.000)
p4@checkpoint: (-0.001, 0.004, -0.006, 0.008)
raw GEN: n/a
raw SIM: SimTrack #835 key=214244

gamma (22)
Particle 291
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.002, -0.004, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.006276
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.006276
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-5.913, 123.158, -217.812, 0.000)
p4@checkpoint: (-0.002, 0.002, -0.004, 0.005)
raw GEN: n/a
raw SIM: SimTrack #836 key=214248

gamma (22)
Particle 292
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.002129
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.002129
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-5.832, 123.162, -217.635, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.002, 0.002)
raw GEN: n/a
raw SIM: SimTrack #837 key=214249

gamma (22)
Particle 293
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.006, 0.005, -0.013, 0.015)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.011319
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.011319
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-5.929, 123.157, -217.699, 0.000)
p4@checkpoint: (-0.006, 0.005, -0.013, 0.015)
raw GEN: n/a
raw SIM: SimTrack #838 key=214250

gamma (22)
Particle 294
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.004, 0.003, -0.009, 0.010)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 1 simE=0.000681
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 2 simE=0.001525
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (5.318, 123.185, -204.467, 0.000)
p4@checkpoint: (0.000, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #839 key=214256

Bremsstrahlung
Vertex 189
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (1.395, 116.065, -202.617, 8.343)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #303 key=238

gamma (22)
Particle 295
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.031, 0.024, -0.062, 0.073)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.064068
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.064068
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-7.105, 123.095, -219.168, 0.000)
p4@checkpoint: (-0.031, 0.024, -0.062, 0.073)
raw GEN: n/a
raw SIM: SimTrack #840 key=214257

gamma (22)
Particle 296
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001005
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001005
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-7.243, 123.087, -219.306, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #841 key=214258

gamma (22)
Particle 297
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.073, 0.055, -0.142, 0.169)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 11 simE=0.115355
direct recHits: 0 missing=11 recoE=0.000000
subgraph simHits: 11 simE=0.115355
subgraph recHits: 0 missing=11 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-7.181, 123.091, -219.111, 0.000)
p4@checkpoint: (-0.073, 0.055, -0.142, 0.169)
raw GEN: n/a
raw SIM: SimTrack #842 key=214259

gamma (22)
Particle 298
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.018, 0.014, -0.035, 0.041)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.052311
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.052311
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-7.069, 123.097, -218.921, 0.000)
p4@checkpoint: (-0.018, 0.014, -0.035, 0.041)
raw GEN: n/a
raw SIM: SimTrack #844 key=214263

gamma (22)
Particle 299
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.005, 0.004, -0.009, 0.011)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.010781
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.010781
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-6.914, 123.106, -218.994, 0.000)
p4@checkpoint: (-0.005, 0.004, -0.009, 0.011)
raw GEN: n/a
raw SIM: SimTrack #845 key=214265

gamma (22)
Particle 300
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.073, 0.057, -0.147, 0.174)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 6 simE=0.002929
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #846 key=214266

PairConversion

Vertex 179
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (1.185, 116.825, -202.853, 8.336)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #285 key=220

gamma (22)
Particle 301
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.016, 0.012, -0.032, 0.038)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.028123
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.028123
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-7.061, 123.098, -219.315, 0.000)
p4@checkpoint: (-0.016, 0.012, -0.032, 0.038)
raw GEN: n/a
raw SIM: SimTrack #847 key=214267

gamma (22)
Particle 302
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.017, 0.013, -0.034, 0.041)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.033135
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.033135
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-7.056, 123.098, -219.083, 0.000)
p4@checkpoint: (-0.017, 0.013, -0.034, 0.041)
raw GEN: n/a
raw SIM: SimTrack #849 key=214269

gamma (22)
Particle 303
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000175
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000175
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-7.411, 123.077, -219.573, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #850 key=214271

gamma (22)
Particle 304
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000138
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000138
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-8.341, 123.018, -221.407, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #851 key=214272

gamma (22)
Particle 305
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.182, 0.121, -0.356, 0.418)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 21 simE=0.457810
direct recHits: 0 missing=21 recoE=0.000000
subgraph simHits: 21 simE=0.457810
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-8.299, 123.020, -221.310, 0.000)
p4@checkpoint: (-0.182, 0.121, -0.356, 0.418)
raw GEN: n/a
raw SIM: SimTrack #853 key=214275

gamma (22)
Particle 306
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.016, 0.007, -0.038, 0.042)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.037752
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.037752
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-12.520, 122.663, -235.863, 0.000)
p4@checkpoint: (-0.016, 0.007, -0.038, 0.042)
raw GEN: n/a
raw SIM: SimTrack #854 key=214277

gamma (22)
Particle 307
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000855
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000855
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-1.495, 123.291, -208.711, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #856 key=214280

e- (11)
Particle 308
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.043, 0.034, -0.088, 0.103)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2 simE=0.001353
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #857 key=214476

Bremsstrahlung
Vertex 183
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (0.095, 117.532, -204.953, 8.419)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #293 key=228

Bremsstrahlung
Vertex 184
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-0.096, 117.607, -205.291, 8.432)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #294 key=229

e+ (-11)
Particle 309
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.030, 0.023, -0.059, 0.070)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 4 simE=0.001576
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #858 key=214477

Bremsstrahlung
Vertex 180
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (3.780, 119.265, -220.467, 8.968)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #287 key=222

Annihilation

Vertex 181
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (3.295, 119.205, -235.265, 9.561)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #288 key=223

Bremsstrahlung
Vertex 182
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (1.568, 116.932, -229.470, 9.389)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #291 key=226

gamma (22)
Particle 310
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.001, -0.004, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000691
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000691
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (14.479, 122.447, -262.690, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #860 key=214483

gamma (22)
Particle 311
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (5.103, 123.194, -235.117, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #861 key=214488

gamma (22)
Particle 312
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000374
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000374
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (7.666, 123.061, -236.509, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #864 key=214503

gamma (22)
Particle 313
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.001, 0.001, -0.002, 0.003)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.000049
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (11.511, 122.762, -209.920, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #866 key=214519

Bremsstrahlung
Vertex 185
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-0.698, 117.511, -208.931, 8.563)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #299 key=234

gamma (22)
Particle 314
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001304
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001304
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-14.202, 122.479, -229.567, 0.000)
p4@checkpoint: (-0.001, 0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #867 key=214521

gamma (22)
Particle 315
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000049
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000049
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (1.344, 123.293, -215.098, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #872 key=214635

gamma (22)
Particle 316
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000278
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000278
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (15.819, 122.281, -198.042, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #873 key=214659

gamma (22)
Particle 317
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000364
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000364
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (15.702, 122.296, -215.822, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #874 key=214668

gamma (22)
Particle 318
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000244
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000244
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (5.988, 123.155, -202.771, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #875 key=214676

gamma (22)
Particle 319
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000183
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000183
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 1 dE=0.000082
subgraph tracker simHits: 1 dE=0.000082
checkpointId: 0
x4@checkpoint: (-115.300, -43.691, -198.465, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #876 key=214677

gamma (22)
Particle 320
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000844
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000844
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-14.905, 122.396, -188.282, 0.000)
p4@checkpoint: (-0.001, 0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #877 key=214745

gamma (22)
Particle 321
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000511
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000511
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-37.141, -117.573, -235.182, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #879 key=215747

gamma (22)
Particle 322
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000343
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000343
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (77.654, 95.774, -218.893, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #880 key=215748

e- (11)
Particle 323
pid: e- (11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.035, 0.075, -0.135, 0.158)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 13 simE=0.091615
direct recHits: 0 missing=13 recoE=0.000000
subgraph simHits: 13 simE=0.091615
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (52.558, 111.537, -195.036, 0.000)
p4@checkpoint: (-0.034, 0.068, -0.132, 0.152)
raw GEN: n/a
raw SIM: SimTrack #881 key=215838

e+ (-11)
Particle 324
pid: e+ (-11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.053, 0.108, -0.188, 0.223)
nProdVtx: 1 nDecayVtx: 12
nParents: 1 nChildren: 12
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 17 simE=0.103475
subgraph recHits: 0 missing=17 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (57.335, 109.158, -194.623, 0.000)
p4@checkpoint: (0.072, 0.015, -0.124, 0.144)
raw GEN: n/a
raw SIM: SimTrack #882 key=215839

Bremsstrahlung
Vertex 192
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (52.593, 105.365, -184.005, 7.583)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #307 key=242

Bremsstrahlung
Vertex 193
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (52.599, 105.377, -184.027, 7.584)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #308 key=243

Bremsstrahlung
Vertex 194
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (52.855, 105.852, -184.959, 7.620)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #309 key=244

Bremsstrahlung
Vertex 195
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (53.318, 106.565, -186.429, 7.676)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #310 key=245

Bremsstrahlung
Vertex 196
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (59.415, 99.197, -220.780, 9.002)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #312 key=247

Bremsstrahlung
Vertex 197
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (56.894, 100.697, -226.809, 9.231)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #314 key=249

Bremsstrahlung
Vertex 198
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (56.889, 100.718, -226.844, 9.232)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #315 key=250

Bremsstrahlung
Vertex 199
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (56.824, 100.945, -227.168, 9.245)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #316 key=251

Bremsstrahlung
Vertex 200
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (56.823, 100.949, -227.174, 9.246)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #317 key=252

Bremsstrahlung
Vertex 201
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (56.707, 101.213, -227.516, 9.261)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #318 key=253

Annihilation

Vertex 202
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (56.789, 103.716, -245.575, 10.251)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #321 key=256

Annihilation

Vertex 203
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (59.607, 99.331, -221.080, 9.217)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #326 key=261

gamma (22)
Particle 325
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.014, 0.029, -0.053, 0.062)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.081207
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.081207
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (55.003, 110.352, -192.965, 0.000)
p4@checkpoint: (0.014, 0.029, -0.053, 0.062)
raw GEN: n/a
raw SIM: SimTrack #883 key=215840

gamma (22)
Particle 326
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000213
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000213
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (55.109, 110.299, -193.551, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #884 key=215841

gamma (22)
Particle 327
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.004455
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.004455
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (55.366, 110.170, -193.503, 0.000)
p4@checkpoint: (0.001, 0.001, -0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #885 key=215842

gamma (22)
Particle 328
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, 0.003, -0.007, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.005296
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.005296
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (55.811, 109.946, -193.683, 0.000)
p4@checkpoint: (0.002, 0.003, -0.007, 0.008)
raw GEN: n/a
raw SIM: SimTrack #886 key=215846

gamma (22)
Particle 329
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.000, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000795
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000795
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.264, -2.983, -251.126, 0.000)
p4@checkpoint: (0.000, -0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #888 key=215856

gamma (22)
Particle 330
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001796
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001796
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (54.294, 110.702, -245.135, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #890 key=215861

gamma (22)
Particle 331
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.003, -0.004, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003510
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003510
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (58.595, 108.487, -236.582, 0.000)
p4@checkpoint: (0.001, 0.002, -0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #891 key=215862

gamma (22)
Particle 332
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.004849
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.004849
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (54.209, 110.744, -242.235, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #892 key=215867

gamma (22)
Particle 333
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000174
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000174
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-119.461, 30.529, -279.791, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #893 key=215868

gamma (22)
Particle 334
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.002, -0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000373
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000373
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (51.982, 111.807, -237.611, 0.000)
p4@checkpoint: (-0.001, 0.002, -0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #894 key=215872

gamma (22)
Particle 335
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000355
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000355
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (47.030, 113.979, -258.948, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #897 key=215889

gamma (22)
Particle 336
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000454
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000454
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (72.276, 99.895, -230.882, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #903 key=215937

gamma (22)
Particle 337
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.145, 1.347, -3.223, 3.676)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 61 simE=3.600389
direct recHits: 0 missing=61 recoE=0.000000
subgraph simHits: 61 simE=3.600389
subgraph recHits: 0 missing=61 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (78.500, 95.082, -220.359, 0.000)
p4@checkpoint: (1.145, 1.347, -3.223, 3.676)
raw GEN: n/a
raw SIM: SimTrack #905 key=217193

gamma (22)
Particle 338
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.079, 0.079, -0.281, 0.303)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=0.001136
direct recHits: 9 missing=0 recoE=0.130594
subgraph simHits: 9 simE=0.001136
subgraph recHits: 9 missing=0 recoE=0.130594
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (84.108, 87.341, -293.500, 0.000)
p4@checkpoint: (0.079, 0.079, -0.281, 0.303)
raw GEN: n/a
raw SIM: SimTrack #906 key=217194

gamma (22)
Particle 339
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.129, 0.146, -0.336, 0.388)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 7 simE=0.001719
subgraph recHits: 6 missing=1 recoE=0.210858
direct tracker simHits: 12 dE=0.001244
subgraph tracker simHits: 24 dE=0.003260
raw GEN: n/a
raw SIM: SimTrack #907 key=219594

PairConversion

Vertex 214
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.666, 54.599, -118.992, 4.925)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #339 key=274

gamma (22)
Particle 340
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.249, 2.053, -3.734, 4.441)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 53 simE=4.288117
subgraph recHits: 0 missing=53 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #908 key=219595

PairConversion

Vertex 206
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (60.097, 98.927, -178.064, 7.379)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #330 key=265

e- (11)
Particle 341
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.379, 0.622, -1.133, 1.347)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 5
nCheckpoints: 1
direct simHits: 30 simE=1.300471
direct recHits: 0 missing=30 recoE=0.000000
subgraph simHits: 31 simE=1.323534
subgraph recHits: 0 missing=31 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (63.588, 105.638, -189.596, 0.000)
p4@checkpoint: (0.294, 0.654, -1.092, 1.306)
raw GEN: n/a
raw SIM: SimTrack #909 key=219596

Bremsstrahlung
Vertex 209
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (60.130, 98.982, -178.162, 7.383)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #334 key=269

Bremsstrahlung
Vertex 210
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (60.515, 99.636, -179.326, 7.430)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #335 key=270

Bremsstrahlung
Vertex 211
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (60.570, 99.731, -179.495, 7.436)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #336 key=271

Bremsstrahlung
Vertex 212
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (61.063, 100.608, -181.030, 7.497)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #337 key=272

Bremsstrahlung
Vertex 213
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (61.064, 100.609, -181.032, 7.498)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #338 key=273

e+ (-11)
Particle 342
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.871, 1.430, -2.601, 3.094)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 38 simE=2.902779
direct recHits: 0 missing=38 recoE=0.000000
subgraph simHits: 38 simE=2.964583
subgraph recHits: 0 missing=38 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (64.183, 105.278, -189.676, 0.000)
p4@checkpoint: (0.925, 1.364, -2.530, 3.020)
raw GEN: n/a
raw SIM: SimTrack #910 key=219597

Bremsstrahlung
Vertex 207
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (60.247, 99.175, -178.511, 7.397)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #332 key=267

Bremsstrahlung
Vertex 208
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (61.117, 100.577, -181.056, 7.498)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #333 key=268

gamma (22)
Particle 343
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.016, 0.027, -0.048, 0.057)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.060493
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.060493
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (64.021, 105.377, -189.718, 0.000)
p4@checkpoint: (0.016, 0.027, -0.048, 0.057)
raw GEN: n/a
raw SIM: SimTrack #912 key=219601

gamma (22)
Particle 344
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001311
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001311
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (64.099, 105.329, -189.701, 0.000)
p4@checkpoint: (0.000, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #913 key=219603

gamma (22)
Particle 345
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.005, 0.008, -0.014, 0.017)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.015886
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.015886
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (64.007, 105.385, -189.711, 0.000)
p4@checkpoint: (0.005, 0.008, -0.014, 0.017)
raw GEN: n/a
raw SIM: SimTrack #914 key=220520

gamma (22)
Particle 346
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.002, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002217
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002217
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (63.864, 105.471, -189.589, 0.000)
p4@checkpoint: (0.001, 0.002, -0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #915 key=220522

gamma (22)
Particle 347
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, -0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002808
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002808
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (63.854, 105.478, -189.561, 0.000)
p4@checkpoint: (0.001, 0.001, -0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #916 key=220524

gamma (22)
Particle 348
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001255
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001255
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (63.781, 105.522, -189.605, 0.000)
p4@checkpoint: (0.000, 0.001, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #917 key=220525

gamma (22)
Particle 349
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000897
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000897
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (61.242, 107.015, -187.039, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #918 key=220526

e- (11)
Particle 350
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.098, 0.111, -0.253, 0.293)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 6 simE=0.001259
direct recHits: 6 missing=0 recoE=0.210858
subgraph simHits: 6 simE=0.001259
subgraph recHits: 6 missing=0 recoE=0.210858
direct tracker simHits: 12 dE=0.001498
subgraph tracker simHits: 12 dE=0.002016
checkpointId: 0
x4@checkpoint: (43.619, 73.919, -293.500, 0.000)
p4@checkpoint: (-0.113, 0.071, -0.239, 0.274)
raw GEN: n/a
raw SIM: SimTrack #919 key=220950

Ionisation

Vertex 215
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (30.064, 52.260, -224.598, 8.976)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #342 key=277

e+ (-11)
Particle 351
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.032, 0.036, -0.083, 0.096)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000461
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000461
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (47.571, 50.286, -293.500, 0.000)
p4@checkpoint: (-0.026, 0.021, -0.052, 0.062)
raw GEN: n/a
raw SIM: SimTrack #920 key=220951

e- (11)
Particle 352
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.003, -0.004, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 2 dE=0.000518
subgraph tracker simHits: 2 dE=0.000518
checkpointId: 0
x4@checkpoint: (31.254, 52.833, -293.500, 0.000)
p4@checkpoint: (0.000, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #923 key=221029

pi+ (211)
Particle 353
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.088, 0.130, -0.129, 0.247)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 15 dE=0.001450
subgraph tracker simHits: 15 dE=0.001856
raw GEN: n/a
raw SIM: SimTrack #926 key=231036

Decay
Vertex 245
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (28.795, -4.110, -48.117, 3.075)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #378 key=313

pdg(221)
Particle 354
pid: pdg(221)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.105, 0.626, -0.819, 1.172)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 62 simE=0.989189
subgraph recHits: 0 missing=62 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 7 dE=0.001311
raw GEN: n/a
raw SIM: SimTrack #927 key=231038

Decay
Vertex 217
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.462, 9.173, -14.775, 0.902)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #346 key=281

gamma (22)
Particle 355
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.193, 0.196, -0.467, 0.542)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 19 simE=0.542647
direct recHits: 0 missing=19 recoE=0.000000
subgraph simHits: 19 simE=0.542647
subgraph recHits: 0 missing=19 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-78.693, 94.922, -218.799, 0.000)
p4@checkpoint: (-0.193, 0.196, -0.467, 0.542)
raw GEN: n/a
raw SIM: SimTrack #928 key=231039

gamma (22)
Particle 356
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.298, 0.430, -0.352, 0.631)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 43 simE=0.446543
subgraph recHits: 0 missing=43 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 7 dE=0.001311
raw GEN: n/a
raw SIM: SimTrack #929 key=231040

PairConversion

Vertex 218
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (65.085, 95.358, -85.384, 5.117)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #347 key=282

Bremsstrahlung
Vertex 238
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (65.141, 95.435, -85.457, 5.121)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #371 key=306

Bremsstrahlung
Vertex 239
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (65.166, 95.480, -85.501, 5.124)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #372 key=307

Bremsstrahlung
Vertex 240
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (65.705, 96.353, -86.297, 5.167)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #373 key=308

e+ (-11)
Particle 357
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.283, 0.406, -0.332, 0.596)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 5
nCheckpoints: 1
direct simHits: 19 simE=0.345647
direct recHits: 0 missing=19 recoE=0.000000
subgraph simHits: 37 simE=0.425951
subgraph recHits: 0 missing=37 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 6 dE=0.001184
checkpointId: 0
x4@checkpoint: (70.657, 101.047, -91.167, 0.000)
p4@checkpoint: (0.235, 0.174, -0.212, 0.361)
raw GEN: n/a
raw SIM: SimTrack #930 key=231042

Bremsstrahlung
Vertex 219
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (65.137, 95.432, -85.445, 5.121)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #348 key=283

Bremsstrahlung
Vertex 220
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (65.182, 95.497, -85.499, 5.124)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #349 key=284

Bremsstrahlung
Vertex 221
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (65.247, 95.590, -85.576, 5.129)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #351 key=286

Bremsstrahlung
Vertex 222
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (65.787, 96.329, -86.233, 5.166)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #352 key=287

Bremsstrahlung
Vertex 223
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (65.821, 96.372, -86.272, 5.169)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #353 key=288

gamma (22)
Particle 358
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001175
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001175
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (69.507, 101.841, -90.698, 0.000)
p4@checkpoint: (0.001, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #931 key=231044

gamma (22)
Particle 359
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.103, 0.147, -0.120, 0.216)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 17 simE=0.074141
subgraph recHits: 0 missing=17 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 6 dE=0.001184
raw GEN: n/a
raw SIM: SimTrack #932 key=231045

PairConversion

Vertex 224
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (65.266, 95.618, -85.597, 5.130)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #354 key=289

Bremsstrahlung
Vertex 235
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (65.268, 95.620, -85.599, 5.130)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #368 key=303

Bremsstrahlung
Vertex 236
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (65.308, 95.937, -85.877, 5.145)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #369 key=304

Bremsstrahlung
Vertex 237
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (64.473, 96.176, -86.639, 5.188)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #370 key=305

gamma (22)
Particle 360
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001020
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001020
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (69.573, 101.797, -90.860, 0.000)
p4@checkpoint: (0.000, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #934 key=231048

gamma (22)
Particle 361
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, 0.002, -0.002, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003647
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003647
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (69.880, 101.586, -91.068, 0.000)
p4@checkpoint: (0.002, 0.002, -0.002, 0.004)
raw GEN: n/a
raw SIM: SimTrack #935 key=231052

gamma (22)
Particle 362
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000320
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000320
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (69.895, 101.576, -91.053, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #936 key=231053

e+ (-11)
Particle 363
pid: e+ (-11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.084, 0.119, -0.097, 0.175)
nProdVtx: 1 nDecayVtx: 10
nParents: 1 nChildren: 10
nCheckpoints: 1
direct simHits: 2 simE=0.008009
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 16 simE=0.033313
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (71.436, 100.498, -90.903, 0.000)
p4@checkpoint: (0.119, 0.040, -0.082, 0.150)
raw GEN: n/a
raw SIM: SimTrack #937 key=231158

Bremsstrahlung
Vertex 225
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (65.585, 96.050, -85.945, 5.151)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #355 key=290

Bremsstrahlung
Vertex 226
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (67.526, 98.200, -87.902, 5.268)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #356 key=291

Bremsstrahlung
Vertex 227
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (75.926, 89.964, -113.124, 6.725)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #358 key=293

Bremsstrahlung
Vertex 228
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (75.966, 89.983, -113.152, 6.727)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #359 key=294

Bremsstrahlung
Vertex 229
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (1.892, 115.769, -191.752, 10.388)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #361 key=296

Bremsstrahlung
Vertex 230
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (1.861, 115.826, -191.919, 10.394)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #362 key=297

Annihilation

Vertex 231
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (1.602, 115.875, -192.126, 10.403)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #363 key=298

Bremsstrahlung
Vertex 232
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (1.795, 116.019, -191.877, 10.396)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #364 key=299

Bremsstrahlung
Vertex 233
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (1.694, 116.575, -192.366, 10.421)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #365 key=300

Annihilation

Vertex 234
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (46.980, -106.677, -200.793, 13.694)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #367 key=302

gamma (22)
Particle 364
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.006, 0.008, -0.006, 0.011)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.013248
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.013248
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (69.814, 101.631, -90.472, 0.000)
p4@checkpoint: (0.006, 0.008, -0.006, 0.011)
raw GEN: n/a
raw SIM: SimTrack #938 key=231160

gamma (22)
Particle 365
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.004, 0.003, -0.003, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.005826
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.005826
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (70.714, 101.007, -90.741, 0.000)
p4@checkpoint: (0.004, 0.003, -0.003, 0.006)
raw GEN: n/a
raw SIM: SimTrack #939 key=231162

gamma (22)
Particle 366
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001329
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001329
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (80.456, 93.433, -115.832, 0.000)
p4@checkpoint: (0.001, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #941 key=231183

gamma (22)
Particle 367
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000096
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000096
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (79.612, 94.153, -117.846, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #942 key=231185

gamma (22)
Particle 368
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000728
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000728
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-1.537, 123.290, -204.904, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #944 key=231206

gamma (22)
Particle 369
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.007, 0.007)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001349
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001349
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-2.289, 123.279, -227.168, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.007, 0.007)
raw GEN: n/a
raw SIM: SimTrack #945 key=231210

gamma (22)
Particle 370
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, -0.000, -0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001748
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001748
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-82.991, 91.188, -221.602, 0.000)
p4@checkpoint: (-0.001, -0.001, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #947 key=231214

gamma (22)
Particle 371
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000063
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000063
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (23.770, 120.987, -199.822, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #948 key=231222

gamma (22)
Particle 372
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000777
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000777
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-1.596, 123.290, -197.575, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #949 key=231223

gamma (22)
Particle 373
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000142
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000142
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (38.958, -116.984, -158.313, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #951 key=231243

gamma (22)
Particle 374
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.015, 0.022, -0.018, 0.032)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.040690
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.040690
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (69.452, 101.879, -90.789, 0.000)
p4@checkpoint: (0.015, 0.022, -0.018, 0.032)
raw GEN: n/a
raw SIM: SimTrack #952 key=231261

gamma (22)
Particle 375
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000138
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000138
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (65.126, 104.697, -94.167, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #953 key=231262

gamma (22)
Particle 376
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, -0.002, -0.003, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 6 dE=0.001184
subgraph tracker simHits: 6 dE=0.001184
checkpointId: 0
x4@checkpoint: (-48.419, -47.148, -293.500, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #954 key=231263

gamma (22)
Particle 377
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000144
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000144
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 1 dE=0.000127
subgraph tracker simHits: 1 dE=0.000127
checkpointId: 0
x4@checkpoint: (92.135, 81.939, -127.023, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #955 key=231279

gamma (22)
Particle 378
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, 0.003, -0.003, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.004898
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.004898
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (69.286, 101.992, -91.913, 0.000)
p4@checkpoint: (0.002, 0.003, -0.003, 0.005)
raw GEN: n/a
raw SIM: SimTrack #956 key=231280

gamma (22)
Particle 379
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.004, 0.010, -0.013, 0.017)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.015550
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.015550
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (68.017, 102.842, -95.012, 0.000)
p4@checkpoint: (0.004, 0.010, -0.013, 0.017)
raw GEN: n/a
raw SIM: SimTrack #957 key=231285

gamma (22)
Particle 380
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001387
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001387
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-113.815, 47.425, 3.694, 0.000)
p4@checkpoint: (-0.001, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #959 key=231474

gamma (22)
Particle 381
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000550
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000550
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-47.508, 113.780, -21.042, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #960 key=231475

gamma (22)
Particle 382
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000176
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000176
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (105.631, 63.599, -67.076, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #961 key=231476

gamma (22)
Particle 383
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.003, 0.002, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001855
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001855
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (86.157, 88.204, 87.460, 0.000)
p4@checkpoint: (-0.001, 0.003, 0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #962 key=231485

n (2112)
Particle 384
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.046, 0.011, 0.026, 0.941)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 14 simE=0.000152
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 14 simE=0.000152
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (108.840, 57.938, 75.635, 0.000)
p4@checkpoint: (0.046, 0.011, 0.026, 0.941)
raw GEN: n/a
raw SIM: SimTrack #963 key=231486

gamma (22)
Particle 385
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000674
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000674
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-108.586, 58.413, 107.951, 0.000)
p4@checkpoint: (-0.001, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #964 key=231488

gamma (22)
Particle 386
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, -0.002, -0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000470
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000470
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-69.616, -101.767, -79.064, 0.000)
p4@checkpoint: (-0.002, -0.002, -0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #965 key=231495

gamma (22)
Particle 387
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, 0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001575
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001575
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (81.962, -92.115, 7.238, 0.000)
p4@checkpoint: (0.001, -0.001, 0.000, 0.002)
raw GEN: n/a
raw SIM: SimTrack #966 key=231496

gamma (22)
Particle 388
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.002, -0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000908
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000908
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-35.372, 118.117, -63.129, 0.000)
p4@checkpoint: (-0.001, 0.002, -0.000, 0.002)
raw GEN: n/a
raw SIM: SimTrack #968 key=231505

mu+ (-13)
Particle 389
caloBoundary
pid: mu+ (-13)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.064, -0.109, -0.129, 0.209)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 2 dE=0.000406
subgraph tracker simHits: 2 dE=0.000406
checkpointId: 0
x4@checkpoint: (13.154, -1.190, -293.500, 0.000)
p4@checkpoint: (-0.000, 0.065, -0.077, 0.146)
raw GEN: n/a
raw SIM: SimTrack #969 key=231514

e- (11)
Particle 390
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, -0.004, -0.011, 0.012)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #970 key=233028

gamma (22)
Particle 391
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.064, 0.022, -0.091, 0.114)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 14 simE=0.113598
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 14 simE=0.113598
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (116.885, 39.254, -162.270, 0.000)
p4@checkpoint: (0.064, 0.022, -0.091, 0.114)
raw GEN: n/a
raw SIM: SimTrack #971 key=258338

gamma (22)
Particle 392
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (11.672, 3.924, -16.550, 20.629)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 119 simE=20.282322
direct recHits: 0 missing=119 recoE=0.000000
subgraph simHits: 119 simE=20.282322
subgraph recHits: 0 missing=119 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (116.885, 39.253, -162.274, 0.000)
p4@checkpoint: (11.672, 3.924, -16.550, 20.629)
raw GEN: n/a
raw SIM: SimTrack #972 key=258339

gamma (22)
Particle 393
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.042, 0.014, -0.060, 0.075)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.072747
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.072747
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (116.882, 39.262, -162.088, 0.000)
p4@checkpoint: (0.042, 0.014, -0.060, 0.075)
raw GEN: n/a
raw SIM: SimTrack #973 key=258340

gamma (22)
Particle 394
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.039, 0.014, -0.056, 0.069)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 17 dE=0.003658
subgraph tracker simHits: 17 dE=0.004239
raw GEN: n/a
raw SIM: SimTrack #974 key=258341

PairConversion

Vertex 291
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (79.741, 28.940, -110.665, 4.941)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #441 key=376

Annihilation

Vertex 292
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (79.753, 28.221, -123.050, 7.576)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #443 key=378

gamma (22)
Particle 395
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.308, 0.135, -0.453, 0.564)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 21 simE=0.530719
direct recHits: 1 missing=20 recoE=0.005203
subgraph simHits: 21 simE=0.530719
subgraph recHits: 1 missing=20 recoE=0.005203
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (113.609, 47.915, -162.603, 0.000)
p4@checkpoint: (0.308, 0.135, -0.453, 0.564)
raw GEN: n/a
raw SIM: SimTrack #975 key=258343

gamma (22)
Particle 396
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000561
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000561
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (108.860, 57.899, -163.606, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #976 key=258346

gamma (22)
Particle 397
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.020, 0.762, -1.722, 2.141)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 89 simE=1.567474
subgraph recHits: 0 missing=89 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 22 dE=0.004528
raw GEN: n/a
raw SIM: SimTrack #977 key=258348

PairConversion

Vertex 256
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (47.714, 22.896, -68.302, 3.178)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #391 key=326

gamma (22)
Particle 398
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000170
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000170
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (27.210, 120.260, -150.333, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #979 key=258353

gamma (22)
Particle 399
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, 0.015, -0.022, 0.026)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000098
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000098
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (32.576, 110.988, -293.500, 0.000)
p4@checkpoint: (-0.003, 0.015, -0.022, 0.026)
raw GEN: n/a
raw SIM: SimTrack #981 key=258359

e- (11)
Particle 400
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.921, 0.688, -1.555, 1.934)
nProdVtx: 1 nDecayVtx: 12
nParents: 1 nChildren: 12
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 89 simE=1.567474
subgraph recHits: 0 missing=89 recoE=0.000000
direct tracker simHits: 7 dE=0.002188
subgraph tracker simHits: 8 dE=0.002222
raw GEN: n/a
raw SIM: SimTrack #982 key=258629

Bremsstrahlung
Vertex 257
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (49.996, 24.658, -72.205, 3.339)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #395 key=330

Bremsstrahlung
Vertex 258
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.091, 77.120, -157.780, 6.888)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #396 key=331

Bremsstrahlung
Vertex 259
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.728, 79.999, -161.762, 7.054)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #397 key=332

Bremsstrahlung
Vertex 260
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.734, 80.025, -161.797, 7.055)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #398 key=333

Bremsstrahlung
Vertex 261
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.746, 80.081, -161.875, 7.058)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #399 key=334

Bremsstrahlung
Vertex 262
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.751, 80.107, -161.911, 7.060)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #400 key=335

Bremsstrahlung
Vertex 263
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.756, 80.129, -161.942, 7.061)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #401 key=336

Bremsstrahlung
Vertex 264
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.763, 80.160, -161.987, 7.063)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #402 key=337

Bremsstrahlung
Vertex 265
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.857, 80.572, -162.569, 7.087)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #403 key=338

Bremsstrahlung
Vertex 266
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.230, 83.468, -166.554, 7.252)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #404 key=339

Bremsstrahlung
Vertex 267
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.261, 84.434, -167.869, 7.306)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #405 key=340

Bremsstrahlung
Vertex 268
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.261, 84.440, -167.877, 7.307)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #406 key=341

e+ (-11)
Particle 401
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.099, 0.074, -0.167, 0.207)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 15 dE=0.002306
subgraph tracker simHits: 15 dE=0.002306
checkpointId: 0
x4@checkpoint: (51.188, 18.593, -293.500, 0.000)
p4@checkpoint: (0.059, 0.036, -0.094, 0.117)
raw GEN: n/a
raw SIM: SimTrack #983 key=258630

gamma (22)
Particle 402
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.066, 0.052, -0.114, 0.142)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 14 simE=0.017099
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #987 key=258837

PairConversion

Vertex 281
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (98.219, 62.921, -155.742, 6.801)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #423 key=358

gamma (22)
Particle 403
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.008, 0.036, -0.050, 0.062)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 12 simE=0.051995
direct recHits: 0 missing=12 recoE=0.000000
subgraph simHits: 12 simE=0.051995
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (85.772, 88.578, -173.638, 0.000)
p4@checkpoint: (0.008, 0.036, -0.050, 0.062)
raw GEN: n/a
raw SIM: SimTrack #988 key=258844

gamma (22)
Particle 404
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.032, 0.145, -0.201, 0.249)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 29 simE=0.206307
subgraph recHits: 0 missing=29 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 1 dE=0.000035
raw GEN: n/a
raw SIM: SimTrack #989 key=258847

PairConversion

Vertex 273
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.731, 80.010, -161.777, 7.054)
nIn: 1 nOut: 1
raw GEN: n/a

gamma (22) raw SIM: SimVertex #415 key=350
Particle 405
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.061, 0.278, -0.385, 0.479)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 28 simE=0.457160
direct recHits: 0 missing=28 recoE=0.000000
subgraph simHits: 28 simE=0.457160
subgraph recHits: 0 missing=28 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (85.636, 88.709, -173.838, 0.000)
p4@checkpoint: (0.061, 0.278, -0.385, 0.479)
raw GEN: n/a
raw SIM: SimTrack #990 key=258848

gamma (22)
Particle 406
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.050, 0.242, -0.338, 0.419)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 17 simE=0.406000
direct recHits: 0 missing=17 recoE=0.000000
subgraph simHits: 17 simE=0.406000
subgraph recHits: 0 missing=17 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (85.549, 88.794, -174.038, 0.000)
p4@checkpoint: (0.050, 0.242, -0.338, 0.419)
raw GEN: n/a
raw SIM: SimTrack #991 key=258849

gamma (22)
Particle 407
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000132
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000132
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (92.549, 81.471, -177.022, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #992 key=258850

gamma (22)
Particle 408
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.009, 0.040, -0.055, 0.068)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 4 simE=0.002558
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #993 key=258851

PairConversion

Vertex 269
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.003, 81.217, -163.452, 7.124)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #410 key=345

gamma (22)
Particle 409
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, 0.011, -0.015, 0.018)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.018431
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.018431
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (85.676, 88.671, -173.953, 0.000)
p4@checkpoint: (0.002, 0.011, -0.015, 0.018)
raw GEN: n/a
raw SIM: SimTrack #994 key=258853

gamma (22)
Particle 410
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000147
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000147
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (85.630, 88.715, -174.118, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #995 key=258856

gamma (22)
Particle 411
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.007, -0.010, 0.012)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.012026
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.012026
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (84.583, 89.714, -175.065, 0.000)
p4@checkpoint: (0.000, 0.007, -0.010, 0.012)
raw GEN: n/a
raw SIM: SimTrack #996 key=258866

gamma (22)
Particle 412
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000125
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000125
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (84.298, 89.982, -175.427, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #997 key=258867

gamma (22)
Particle 413
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.004, 0.253, -0.331, 0.417)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 21 simE=0.395492
direct recHits: 0 missing=21 recoE=0.000000
subgraph simHits: 21 simE=0.395492
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (84.344, 89.939, -175.075, 0.000)
p4@checkpoint: (0.004, 0.253, -0.331, 0.417)
raw GEN: n/a
raw SIM: SimTrack #998 key=258868

e+ (-11)
Particle 414
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.008, 0.036, -0.049, 0.061)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 4 simE=0.002558
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1002 key=259186

Bremsstrahlung
Vertex 270
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.392, 82.156, -164.834, 7.181)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #411 key=346

Bremsstrahlung
Vertex 271
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (86.899, 78.140, -191.353, 8.486)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #413 key=348

Bremsstrahlung
Vertex 272
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (87.652, 77.293, -191.491, 8.531)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #414 key=349

gamma (22)
Particle 415
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001105
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001105
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (87.603, 86.768, -172.192, 0.000)
p4@checkpoint: (0.000, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1003 key=259187

gamma (22)
Particle 416
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000740
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000740
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (93.982, 79.814, -194.696, 0.000)
p4@checkpoint: (0.001, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1005 key=259207

gamma (22)
Particle 417
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000713
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000713
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-22.544, -121.222, -233.593, 0.000)
p4@checkpoint: (-0.000, -0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1006 key=259208

e+ (-11)
Particle 418
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.032, 0.140, -0.195, 0.242)
nProdVtx: 1 nDecayVtx: 7
nParents: 1 nChildren: 7
nCheckpoints: 1
direct simHits: 20 simE=0.175951
direct recHits: 0 missing=20 recoE=0.000000
subgraph simHits: 29 simE=0.206307
subgraph recHits: 0 missing=29 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 1 dE=0.000035
checkpointId: 0
x4@checkpoint: (87.692, 86.677, -171.212, 0.000)
p4@checkpoint: (0.093, 0.076, -0.148, 0.191)
raw GEN: n/a
raw SIM: SimTrack #1007 key=259778

Bremsstrahlung
Vertex 274
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.745, 80.072, -161.863, 7.058)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #416 key=351

Bremsstrahlung
Vertex 275
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.752, 80.104, -161.908, 7.060)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #417 key=352

Bremsstrahlung
Vertex 276
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.813, 83.026, -165.515, 7.219)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #418 key=353

Bremsstrahlung
Vertex 277
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (85.063, 83.514, -166.138, 7.246)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #419 key=354

Bremsstrahlung
Vertex 278
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.986, 80.875, -163.404, 7.122)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #420 key=355

Bremsstrahlung
Vertex 279
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.829, 80.031, -162.017, 7.063)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #421 key=356

Annihilation

Vertex 280
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.834, 80.024, -162.019, 7.064)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #422 key=357

gamma (22)
Particle 419
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, 0.008, -0.012, 0.014)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.014270
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.014270
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (85.755, 88.595, -173.868, 0.000)
p4@checkpoint: (0.002, 0.008, -0.012, 0.014)
raw GEN: n/a
raw SIM: SimTrack #1008 key=259780

gamma (22)
Particle 420
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000626
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000626
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (76.715, 96.529, -169.209, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1009 key=259781

gamma (22)
Particle 421
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000245
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000245
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (86.943, 87.429, -170.958, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1010 key=259786

gamma (22)
Particle 422
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.004, 0.008, -0.011, 0.015)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.014734
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.014734
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (87.092, 87.281, -171.278, 0.000)
p4@checkpoint: (0.004, 0.008, -0.011, 0.015)
raw GEN: n/a
raw SIM: SimTrack #1011 key=259787

gamma (22)
Particle 423
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000105
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000105
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (89.133, 85.195, -176.246, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1012 key=259928

gamma (22)
Particle 424
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 1 dE=0.000035
subgraph tracker simHits: 1 dE=0.000035
checkpointId: 0
x4@checkpoint: (27.478, -88.274, -293.500, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1013 key=259944

gamma (22)
Particle 425
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000375
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000375
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (89.897, 84.388, -156.883, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1014 key=259946

e- (11)
Particle 426
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.039, 0.033, -0.068, 0.085)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 6 simE=0.011239
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1015 key=260000

Bremsstrahlung
Vertex 287
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (98.450, 63.133, -156.119, 6.817)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #433 key=368

Bremsstrahlung
Vertex 288
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (95.881, 65.231, -175.205, 7.700)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #438 key=373

Bremsstrahlung
Vertex 289
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (96.349, 65.543, -175.812, 7.728)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #439 key=374

Bremsstrahlung
Vertex 290
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (94.241, 67.403, -179.135, 7.921)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #440 key=375

e+ (-11)
Particle 427
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.027, 0.021, -0.046, 0.057)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 5
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 8 simE=0.005860
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1016 key=260001

Bremsstrahlung
Vertex 282
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (100.816, 63.524, -158.906, 6.940)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #424 key=359

Bremsstrahlung
Vertex 283
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (99.108, 59.711, -170.156, 7.548)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #427 key=362

Bremsstrahlung
Vertex 284
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (99.116, 59.716, -170.158, 7.548)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #428 key=363

Bremsstrahlung
Vertex 285
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (99.200, 59.774, -170.130, 7.551)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #429 key=364

Bremsstrahlung
Vertex 286
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (100.448, 57.397, -169.492, 7.441)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #432 key=367

gamma (22)
Particle 428
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000576
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000576
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (109.340, 56.988, -169.540, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1017 key=260004

gamma (22)
Particle 429
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, 0.002, 0.000, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.003636
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.003636
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (105.013, 64.615, -169.822, 0.000)
p4@checkpoint: (0.003, 0.002, 0.000, 0.004)
raw GEN: n/a
raw SIM: SimTrack #1020 key=260018

gamma (22)
Particle 430
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, 0.000, 0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001123
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001123
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (105.831, 63.267, -168.670, 0.000)
p4@checkpoint: (0.001, 0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1021 key=260019

gamma (22)
Particle 431
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000385
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000385
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (103.473, 67.054, -166.936, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1022 key=260020

gamma (22)
Particle 432
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000139
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000139
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (104.042, 66.168, -176.675, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1025 key=260039

gamma (22)
Particle 433
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, 0.002, -0.004, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.005295
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.005295
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (103.095, 67.634, -163.455, 0.000)
p4@checkpoint: (0.003, 0.002, -0.004, 0.005)
raw GEN: n/a
raw SIM: SimTrack #1026 key=260051

gamma (22)
Particle 434
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, 0.001, -0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.004044
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.004044
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (102.791, 68.095, -184.973, 0.000)
p4@checkpoint: (0.002, 0.001, -0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #1031 key=260072

gamma (22)
Particle 435
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001194
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001194
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (101.441, 70.090, -182.341, 0.000)
p4@checkpoint: (0.001, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1032 key=260074

gamma (22)
Particle 436
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000706
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000706
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-113.447, -48.298, -136.081, 0.000)
p4@checkpoint: (-0.001, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1033 key=260078

e- (11)
Particle 437
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.032, 0.011, -0.046, 0.057)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 2 dE=0.000581
subgraph tracker simHits: 2 dE=0.000581
checkpointId: 0
x4@checkpoint: (78.894, 33.127, -293.500, 0.000)
p4@checkpoint: (-0.010, -0.018, -0.042, 0.046)
raw GEN: n/a
raw SIM: SimTrack #1034 key=260496

gamma (22)
Particle 438
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.300, -0.027, -200.234, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1036 key=260505

pi0 (111)
Particle 439
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.666, -0.763, 1.566, 1.870)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 58 simE=0.907361
subgraph recHits: 25 missing=33 recoE=0.647558
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 40 dE=0.005044
raw GEN: n/a
raw SIM: SimTrack #1037 key=273891

Decay
Vertex 330
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-21.556, -7.392, 33.263, 1.449)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #493 key=428

pi0 (111)
Particle 440
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.248, 0.068, 0.550, 0.622)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 17 simE=0.198947
subgraph recHits: 10 missing=7 recoE=0.245743
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1038 key=273892

Decay
Vertex 329
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-21.556, -7.392, 33.263, 1.449)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #492 key=427

p (2212)
Particle 441
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-2.035, -1.249, 2.386, 3.504)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 15
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 112 simE=0.706666
subgraph recHits: 0 missing=112 recoE=0.000000
direct tracker simHits: 16 dE=0.083660
subgraph tracker simHits: 46 dE=0.098791
raw GEN: n/a
raw SIM: SimTrack #1039 key=273894

HadronInelastic

Vertex 324
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-25.696, -9.868, 38.086, 1.686)
nIn: 1 nOut: 7
raw GEN: n/a
raw SIM: SimVertex #487 key=422

HadronAtRest

Vertex 325
domain: SIM
reason: HadronAtRest
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-22.108, -10.440, 33.619, 2.566)
nIn: 1 nOut: 6
raw GEN: n/a
raw SIM: SimVertex #488 key=423

HadronInelastic

Vertex 326
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-17.917, -30.219, 8.910, 10.403)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #489 key=424

pi0 (111)
Particle 442
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-15.809, -4.974, 21.165, 26.882)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 109 simE=26.309044
subgraph recHits: 0 missing=109 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 1 dE=0.000441
raw GEN: n/a
raw SIM: SimTrack #1040 key=273895

Decay
Vertex 309
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-21.557, -7.392, 33.264, 1.449)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #472 key=407

pi- (-211)
Particle 443
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-8.154, -2.762, 10.976, 13.950)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 6
nCheckpoints: 1
direct simHits: 25 simE=0.409700
direct recHits: 1 missing=24 recoE=0.025833
subgraph simHits: 31 simE=0.432584
subgraph recHits: 1 missing=30 recoE=0.025833
direct tracker simHits: 31 dE=0.004131
subgraph tracker simHits: 31 dE=0.004691
checkpointId: 0
x4@checkpoint: (-114.427, -45.927, 161.525, 0.000)
p4@checkpoint: (-7.653, -3.768, 10.866, 13.815)
raw GEN: n/a
raw SIM: SimTrack #1041 key=273896

Ionisation

Vertex 306
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-66.083, -24.053, 93.867, 4.019)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #466 key=401

NuclearCapture
Vertex 307
domain: SIM
reason: NuclearCapture
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (49.495, 105.129, 208.249, 298.969)
nIn: 1 nOut: 4
raw GEN: n/a
raw SIM: SimVertex #467 key=402

Annihilation

Vertex 308
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.604, 90.412, 188.308, 308.390)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #468 key=403

pi0 (111)
Particle 444
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-1.112, -0.661, 1.726, 2.162)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 46 simE=2.030199
subgraph recHits: 0 missing=46 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 1 dE=0.000002
raw GEN: n/a
raw SIM: SimTrack #1042 key=273897

Decay
Vertex 298
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-21.556, -7.392, 33.263, 1.449)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #453 key=388

pi+ (211)
Particle 445
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-5.010, -1.239, 7.373, 9.001)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 123 simE=4.884379
direct recHits: 0 missing=123 recoE=0.000000
subgraph simHits: 128 simE=4.892210
subgraph recHits: 0 missing=128 recoE=0.000000
direct tracker simHits: 16 dE=0.002135
subgraph tracker simHits: 16 dE=0.002135
checkpointId: 0
x4@checkpoint: (-121.594, -20.439, 177.642, 0.000)
p4@checkpoint: (-5.135, -0.138, 7.311, 8.936)
raw GEN: n/a
raw SIM: SimTrack #1043 key=273898

HadronInelastic

Vertex 295
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (8.142, -41.115, 106.517, 35.008)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #450 key=385

HadronInelastic

Vertex 296
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (9.130, 85.366, 123.116, 87.295)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #451 key=386

Bremsstrahlung
Vertex 297
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-114.796, -20.830, 179.352, 7.362)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #452 key=387

pi- (-211)
Particle 446
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.311, -0.038, 0.038, 0.345)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.063158
direct recHits: 2 missing=5 recoE=0.052670
subgraph simHits: 7 simE=0.063158
subgraph recHits: 2 missing=5 recoE=0.052670
direct tracker simHits: 24 dE=0.005695
subgraph tracker simHits: 24 dE=0.005695
checkpointId: 0
x4@checkpoint: (-33.798, -43.788, 293.500, 0.000)
p4@checkpoint: (0.079, -0.113, 0.165, 0.256)
raw GEN: n/a
raw SIM: SimTrack #1044 key=273899

pi+ (211)
Particle 447
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-1.008, 0.122, 1.525, 1.838)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 71 simE=0.974408
direct recHits: 1 missing=70 recoE=0.295328
subgraph simHits: 72 simE=0.974772
subgraph recHits: 1 missing=71 recoE=0.295328
direct tracker simHits: 14 dE=0.001668
subgraph tracker simHits: 14 dE=0.001668
checkpointId: 0
x4@checkpoint: (-98.803, 73.762, 214.343, 0.000)
p4@checkpoint: (-0.092, 1.021, 1.389, 1.732)
raw GEN: n/a
raw SIM: SimTrack #1045 key=273900

Bremsstrahlung
Vertex 294
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-99.887, 64.346, 218.197, 8.877)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #445 key=380

gamma (22)
Particle 448
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.002, 0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.002849
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.002849
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (55.129, -110.289, 78.336, 0.000)
p4@checkpoint: (0.002, -0.002, 0.001, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1046 key=273902

gamma (22)
Particle 449
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, 0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000364
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000364
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-106.211, 62.626, 225.759, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1047 key=274244

gamma (22)
Particle 450
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.003, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003710
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003710
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-8.923, -122.977, 170.345, 0.000)
p4@checkpoint: (-0.001, -0.003, 0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1052 key=275346

gamma (22)
Particle 451
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002892
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002892
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (119.641, 29.816, 1.049, 0.000)
p4@checkpoint: (0.001, 0.001, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1053 key=275347

gamma (22)
Particle 452
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000165
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000165
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-121.292, -22.164, 271.603, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1054 key=275359

gamma (22)
Particle 453
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001064
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001064
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-121.357, -21.801, 182.728, 0.000)
p4@checkpoint: (-0.001, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1055 key=276128

gamma (22)
Particle 454
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.562, -0.273, 0.901, 1.097)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 17 simE=0.977170
subgraph recHits: 0 missing=17 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 1 dE=0.000002
raw GEN: n/a
raw SIM: SimTrack #1056 key=276132

PairConversion

Vertex 299
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-105.520, -48.119, 167.839, 6.912)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #454 key=389

gamma (22)
Particle 455
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.550, -0.388, 0.825, 1.065)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 29 simE=1.053029
direct recHits: 0 missing=29 recoE=0.000000
subgraph simHits: 29 simE=1.053029
subgraph recHits: 0 missing=29 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-104.302, -65.756, 157.376, 0.000)
p4@checkpoint: (-0.550, -0.388, 0.825, 1.065)
raw GEN: n/a
raw SIM: SimTrack #1057 key=276133

e- (11)
Particle 456
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.052, -0.025, 0.083, 0.101)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2 simE=0.000265
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 1 dE=0.000002
subgraph tracker simHits: 1 dE=0.000002
raw GEN: n/a
raw SIM: SimTrack #1058 key=276460

Bremsstrahlung
Vertex 304
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-101.381, -55.840, 185.240, 7.688)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #460 key=395

Bremsstrahlung
Vertex 305
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-101.257, -55.642, 185.493, 7.699)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #461 key=396

e+ (-11)
Particle 457
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.510, -0.247, 0.818, 0.995)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 15 simE=0.909260
direct recHits: 0 missing=15 recoE=0.000000
subgraph simHits: 15 simE=0.976905
subgraph recHits: 0 missing=15 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-112.371, -50.750, 178.590, 0.000)
p4@checkpoint: (-0.495, -0.145, 0.758, 0.917)
raw GEN: n/a
raw SIM: SimTrack #1059 key=276461

Bremsstrahlung
Vertex 300
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-105.537, -48.128, 167.867, 6.913)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #455 key=390

Bremsstrahlung
Vertex 301
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-105.586, -48.151, 167.945, 6.916)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #456 key=391

Bremsstrahlung
Vertex 302
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-106.073, -48.383, 168.731, 6.948)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #457 key=392

Bremsstrahlung
Vertex 303
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-108.520, -49.434, 172.612, 7.105)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #458 key=393

gamma (22)
Particle 458
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.011, -0.006, 0.018, 0.022)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.027997
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.027997
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-112.118, -51.308, 178.416, 0.000)
p4@checkpoint: (-0.011, -0.006, 0.018, 0.022)
raw GEN: n/a
raw SIM: SimTrack #1060 key=276462

gamma (22)
Particle 459
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000573
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000573
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-112.111, -51.322, 178.480, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1061 key=276463

gamma (22)
Particle 460
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001107
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001107
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-112.135, -51.271, 178.535, 0.000)
p4@checkpoint: (-0.001, -0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1062 key=276465

gamma (22)
Particle 461
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.022, -0.009, 0.035, 0.042)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.037968
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.037968
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-112.294, -50.921, 178.571, 0.000)
p4@checkpoint: (-0.022, -0.009, 0.035, 0.042)
raw GEN: n/a
raw SIM: SimTrack #1063 key=276470

gamma (22)
Particle 462
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000194
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000194
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-113.479, -48.222, 181.938, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1065 key=276788

gamma (22)
Particle 463
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, 0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000071
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000071
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-50.056, 36.621, 293.500, 0.000)
p4@checkpoint: (0.001, 0.001, 0.002, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1066 key=276792

e- (11)
Particle 464
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.051, -0.013, 0.059, 0.079)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 6 dE=0.000559
subgraph tracker simHits: 6 dE=0.000559
checkpointId: 0
x4@checkpoint: (-69.423, -29.035, 293.500, 0.000)
p4@checkpoint: (0.002, -0.032, 0.052, 0.061)
raw GEN: n/a
raw SIM: SimTrack #1071 key=276855

gamma (22)
Particle 465
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002136
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002136
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (44.133, 115.131, 201.441, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1072 key=277057

gamma (22)
Particle 466
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.014860
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.014860
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.595, 13.170, 120.850, 0.000)
p4@checkpoint: (0.001, -0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1073 key=277058

gamma (22)
Particle 467
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003022
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003022
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.523, 114.589, 201.994, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1074 key=277059

gamma (22)
Particle 468
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001127
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001127
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (26.700, 120.375, 187.968, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1075 key=277060

gamma (22)
Particle 469
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001738
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001738
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-101.399, 70.151, 165.473, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1076 key=277067

gamma (22)
Particle 470
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-4.163, -1.372, 5.595, 7.107)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 63 simE=6.889456
subgraph recHits: 0 missing=63 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1080 key=279480

PairConversion

Vertex 319
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-110.367, -36.664, 152.632, 6.507)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #482 key=417

gamma (22)
Particle 471
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-11.646, -3.602, 15.570, 19.775)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 85 simE=19.419586
subgraph recHits: 0 missing=85 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 1 dE=0.000441
raw GEN: n/a
raw SIM: SimTrack #1081 key=279481

PairConversion

Vertex 310
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-110.962, -35.042, 152.794, 6.513)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #473 key=408

e- (11)
Particle 472
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-3.846, -1.192, 5.144, 6.532)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 48 simE=6.343877
subgraph recHits: 0 missing=48 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 1 dE=0.000441
raw GEN: n/a
raw SIM: SimTrack #1082 key=279482

Bremsstrahlung
Vertex 314
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-111.168, -35.105, 153.069, 6.525)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #477 key=412

Bremsstrahlung
Vertex 315
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-112.329, -38.026, 157.771, 6.721)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #478 key=413

Bremsstrahlung
Vertex 316
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-109.388, -37.104, 162.547, 6.934)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #479 key=414

Bremsstrahlung
Vertex 317
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-109.384, -37.100, 162.548, 6.934)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #480 key=415

e+ (-11)
Particle 473
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-7.799, -2.412, 10.427, 13.242)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 55 simE=8.588311
direct recHits: 0 missing=55 recoE=0.000000
subgraph simHits: 71 simE=13.075712
subgraph recHits: 0 missing=71 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-117.602, -37.048, 161.648, 0.000)
p4@checkpoint: (-5.156, -1.517, 6.860, 8.714)
raw GEN: n/a
raw SIM: SimTrack #1083 key=279483

Bremsstrahlung
Vertex 311
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-111.062, -35.073, 152.928, 6.519)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #474 key=409

Bremsstrahlung
Vertex 312
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-111.181, -35.109, 153.086, 6.526)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #475 key=410

Annihilation

Vertex 313
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-101.089, -58.369, 195.697, 8.677)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #476 key=411

gamma (22)
Particle 474
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.184, -0.057, 0.246, 0.313)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 14 simE=0.289953
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 14 simE=0.289953
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-117.589, -37.089, 161.653, 0.000)
p4@checkpoint: (-0.184, -0.057, 0.246, 0.313)
raw GEN: n/a
raw SIM: SimTrack #1084 key=279484

gamma (22)
Particle 475
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-2.479, -0.765, 3.314, 4.208)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 41 simE=4.195853
direct recHits: 0 missing=41 recoE=0.000000
subgraph simHits: 41 simE=4.195853
subgraph recHits: 0 missing=41 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-117.590, -37.088, 161.653, 0.000)
p4@checkpoint: (-2.479, -0.765, 3.314, 4.208)
raw GEN: n/a
raw SIM: SimTrack #1085 key=279486

gamma (22)
Particle 476
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000690
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000690
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-103.018, -67.751, 193.260, 0.000)
p4@checkpoint: (-0.000, -0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1086 key=282177

gamma (22)
Particle 477
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000903
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000903
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-111.982, -51.604, 194.310, 0.000)
p4@checkpoint: (-0.001, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1087 key=282178

gamma (22)
Particle 478
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-3.823, -1.188, 5.112, 6.493)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 45 simE=6.343405
direct recHits: 0 missing=45 recoE=0.000000
subgraph simHits: 45 simE=6.343405
subgraph recHits: 0 missing=45 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-117.586, -37.101, 161.651, 0.000)
p4@checkpoint: (-3.823, -1.188, 5.112, 6.493)
raw GEN: n/a
raw SIM: SimTrack #1088 key=283611

gamma (22)
Particle 479
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000311
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000311
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-111.841, -51.909, 183.560, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1089 key=283614

gamma (22)
Particle 480
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.004, 0.005, 0.002, 0.006)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.000162
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (31.545, -36.772, 293.500, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1090 key=283623

Bremsstrahlung
Vertex 318
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.028, 115.455, 210.664, 13.501)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #481 key=416

gamma (22)
Particle 481
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 1 dE=0.000441
subgraph tracker simHits: 1 dE=0.000441
checkpointId: 0
x4@checkpoint: (-120.230, -18.076, 293.500, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1091 key=283625

gamma (22)
Particle 482
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000162
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000162
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.236, 76.196, 293.500, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1092 key=283641

e- (11)
Particle 483
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.321, -0.106, 0.432, 0.549)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 18 simE=0.527283
direct recHits: 0 missing=18 recoE=0.000000
subgraph simHits: 18 simE=0.529936
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-116.762, -39.617, 161.529, 0.000)
p4@checkpoint: (-0.285, -0.175, 0.423, 0.539)
raw GEN: n/a
raw SIM: SimTrack #1093 key=285608

Bremsstrahlung
Vertex 322
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-110.391, -36.672, 152.664, 6.509)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #485 key=420

Bremsstrahlung
Vertex 323
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-110.991, -36.877, 153.468, 6.543)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #486 key=421

e+ (-11)
Particle 484
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-3.843, -1.265, 5.162, 6.559)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 48 simE=4.518468
direct recHits: 0 missing=48 recoE=0.000000
subgraph simHits: 56 simE=6.359519
subgraph recHits: 0 missing=56 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-117.049, -38.760, 161.575, 0.000)
p4@checkpoint: (-2.731, -0.816, 3.638, 4.621)
raw GEN: n/a
raw SIM: SimTrack #1094 key=285609

Bremsstrahlung
Vertex 320
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-110.493, -36.706, 152.801, 6.515)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #483 key=418

Bremsstrahlung
Vertex 321
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-110.558, -36.727, 152.888, 6.518)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #484 key=419

gamma (22)
Particle 485
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.017, -0.006, 0.023, 0.030)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.029701
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.029701
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-117.020, -38.849, 161.570, 0.000)
p4@checkpoint: (-0.017, -0.006, 0.023, 0.030)
raw GEN: n/a
raw SIM: SimTrack #1095 key=285610

gamma (22)
Particle 486
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-1.113, -0.365, 1.496, 1.900)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 29 simE=1.811350
direct recHits: 0 missing=29 recoE=0.000000
subgraph simHits: 29 simE=1.811350
subgraph recHits: 0 missing=29 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-117.021, -38.846, 161.570, 0.000)
p4@checkpoint: (-1.113, -0.365, 1.496, 1.900)
raw GEN: n/a
raw SIM: SimTrack #1096 key=285611

gamma (22)
Particle 487
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000069
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000069
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-117.012, -38.873, 161.532, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1097 key=287685

gamma (22)
Particle 488
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, -0.001, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002585
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002585
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-116.979, -38.970, 161.468, 0.000)
p4@checkpoint: (-0.002, -0.001, 0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1098 key=287690

p (2212)
Particle 489
caloBoundary
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-1.028, -1.064, 0.919, 1.979)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 46 simE=0.645819
direct recHits: 0 missing=46 recoE=0.000000
subgraph simHits: 46 simE=0.645819
subgraph recHits: 0 missing=46 recoE=0.000000
direct tracker simHits: 13 dE=0.001391
subgraph tracker simHits: 13 dE=0.001391
checkpointId: 0
x4@checkpoint: (-113.154, -48.980, 98.644, 0.000)
p4@checkpoint: (-1.456, -0.067, 0.907, 1.956)
raw GEN: n/a
raw SIM: SimTrack #1099 key=287851

n (2112)
Particle 490
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.355, -0.032, 0.918, 1.361)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 6 simE=0.013241
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1100 key=287852

HadronInelastic

Vertex 328
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (16.048, -13.582, 146.204, 7.031)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #491 key=426

p (2212)
Particle 491
caloBoundary
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.550, 0.170, 0.311, 1.144)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 16 dE=0.010070
subgraph tracker simHits: 16 dE=0.010070
checkpointId: 0
x4@checkpoint: (-13.442, 9.052, 293.500, 0.000)
p4@checkpoint: (-0.237, 0.145, 0.131, 0.987)
raw GEN: n/a
raw SIM: SimTrack #1101 key=287853

p (2212)
Particle 492
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.420, 0.105, -0.413, 1.113)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 4 simE=0.000117
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 6 dE=0.002274
subgraph tracker simHits: 6 dE=0.002274
raw GEN: n/a
raw SIM: SimTrack #1102 key=287854

HadronInelastic

Vertex 327
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-17.388, -0.531, -152.605, 20.682)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #490 key=425

n (2112)
Particle 493
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.299, 0.217, -0.066, 1.012)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=0.010522
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=0.010522
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-111.546, 52.540, 19.197, 0.000)
p4@checkpoint: (-0.299, 0.217, -0.066, 1.012)
raw GEN: n/a
raw SIM: SimTrack #1104 key=287856

n (2112)
Particle 494
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.025, -0.127, 0.024, 0.949)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.008004
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.008004
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-2.794, -123.268, 92.373, 0.000)
p4@checkpoint: (0.042, -0.051, 0.047, 0.943)
raw GEN: n/a
raw SIM: SimTrack #1105 key=287863

gamma (22)
Particle 495
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, 0.001, 0.000, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002160
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002160
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (120.258, 27.218, 45.791, 0.000)
p4@checkpoint: (0.003, 0.001, 0.000, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1107 key=287866

gamma (22)
Particle 496
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000975
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000975
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-105.256, -64.219, 219.758, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1109 key=287937

n (2112)
Particle 497
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.247, 0.112, 0.156, 0.990)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000150
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000150
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-118.541, 33.924, 87.034, 0.000)
p4@checkpoint: (-0.255, 0.118, 0.137, 0.990)
raw GEN: n/a
raw SIM: SimTrack #1110 key=287939

n (2112)
Particle 498
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.115, -0.139, -0.047, 0.958)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 32 simE=0.009082
direct recHits: 0 missing=32 recoE=0.000000
subgraph simHits: 32 simE=0.009082
subgraph recHits: 0 missing=32 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-77.633, -95.791, 60.834, 0.000)
p4@checkpoint: (-0.093, -0.140, 0.046, 0.956)
raw GEN: n/a
raw SIM: SimTrack #1111 key=287940

n (2112)
Particle 499
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.028, 0.031, 0.148, 0.952)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.010962
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.010962
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-13.606, 122.547, 258.696, 0.000)
p4@checkpoint: (0.006, 0.078, 0.131, 0.952)
raw GEN: n/a
raw SIM: SimTrack #1112 key=287941

gamma (22)
Particle 500
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, -0.001, -0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.004161
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.004161
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-114.862, -44.830, -13.546, 0.000)
p4@checkpoint: (-0.003, -0.001, -0.001, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1113 key=287943

gamma (22)
Particle 501
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.001, -0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000478
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000478
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (73.567, -98.948, -154.424, 0.000)
p4@checkpoint: (0.000, -0.001, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1114 key=287944

gamma (22)
Particle 502
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, 0.002, 0.000, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000182
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000182
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 2 dE=0.001396
subgraph tracker simHits: 2 dE=0.001396
checkpointId: 0
x4@checkpoint: (-88.163, 86.199, 24.767, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1115 key=287955

gamma (22)
Particle 503
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000813
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000813
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-110.201, -55.304, 48.149, 0.000)
p4@checkpoint: (-0.001, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1116 key=287957

n (2112)
Particle 504
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.321, -0.091, -0.225, 1.022)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.000117
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.000117
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-119.935, -28.609, -223.446, 0.000)
p4@checkpoint: (-0.368, -0.007, -0.161, 1.022)
raw GEN: n/a
raw SIM: SimTrack #1117 key=288361

n (2112)
Particle 505
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.415, -0.322, 0.341, 1.129)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.012482
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.012482
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (96.892, -76.255, 212.656, 0.000)
p4@checkpoint: (0.415, -0.322, 0.341, 1.129)
raw GEN: n/a
raw SIM: SimTrack #1119 key=288757

gamma (22)
Particle 506
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000759
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000759
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (55.634, 110.035, -43.912, 0.000)
p4@checkpoint: (0.000, 0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1120 key=288759

gamma (22)
Particle 507
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.045, -0.006, -0.016, 0.048)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.056888
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.056888
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-121.529, -20.823, -2.820, 0.000)
p4@checkpoint: (-0.045, -0.006, -0.016, 0.048)
raw GEN: n/a
raw SIM: SimTrack #1121 key=289310

gamma (22)
Particle 508
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.041, -0.086, 0.090, 0.131)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=0.144764
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=0.144764
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (31.484, -119.213, 149.628, 0.000)
p4@checkpoint: (0.041, -0.086, 0.090, 0.131)
raw GEN: n/a
raw SIM: SimTrack #1122 key=289311

gamma (22)
Particle 509
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.134, 0.022, 0.145, 0.199)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.196362
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.196362
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.936, 9.465, 142.906, 0.000)
p4@checkpoint: (-0.134, 0.022, 0.145, 0.199)
raw GEN: n/a
raw SIM: SimTrack #1123 key=289372

gamma (22)
Particle 510
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.114, 0.045, 0.405, 0.423)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 11 simE=0.002585
direct recHits: 10 missing=1 recoE=0.245743
subgraph simHits: 11 simE=0.002585
subgraph recHits: 10 missing=1 recoE=0.245743
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.779, 21.760, 293.500, 0.000)
p4@checkpoint: (-0.114, 0.045, 0.405, 0.423)
raw GEN: n/a
raw SIM: SimTrack #1124 key=289373

gamma (22)
Particle 511
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.309, -0.449, 0.880, 1.035)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 30 simE=0.902805
subgraph recHits: 0 missing=30 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1125 key=290128

PairConversion

Vertex 337
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-76.480, -87.268, 189.835, 7.592)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #502 key=437

gamma (22)
Particle 512
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.358, -0.314, 0.686, 0.835)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 28 simE=0.004556
subgraph recHits: 25 missing=3 recoE=0.647558
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 40 dE=0.005044
raw GEN: n/a
raw SIM: SimTrack #1126 key=290129

PairConversion

Vertex 331
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-58.439, -39.790, 103.961, 4.320)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #494 key=429

e- (11)
Particle 513
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.137, -0.120, 0.262, 0.319)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 2 simE=0.000209
direct recHits: 2 missing=0 recoE=0.034933
subgraph simHits: 10 simE=0.001963
subgraph recHits: 8 missing=2 recoE=0.150689
direct tracker simHits: 14 dE=0.001665
subgraph tracker simHits: 25 dE=0.003369
checkpointId: 0
x4@checkpoint: (-46.713, -61.503, 293.500, 0.000)
p4@checkpoint: (0.001, 0.065, 0.110, 0.128)
raw GEN: n/a
raw SIM: SimTrack #1127 key=290130

Bremsstrahlung
Vertex 334
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-52.943, -66.689, 152.591, 6.286)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #498 key=433

e+ (-11)
Particle 514
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.220, -0.194, 0.424, 0.516)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 11 simE=0.001523
direct recHits: 11 missing=0 recoE=0.251703
subgraph simHits: 18 simE=0.002593
subgraph recHits: 17 missing=1 recoE=0.496868
direct tracker simHits: 15 dE=0.001675
subgraph tracker simHits: 15 dE=0.001675
checkpointId: 0
x4@checkpoint: (-54.515, -14.807, 293.500, 0.000)
p4@checkpoint: (0.025, -0.215, 0.322, 0.388)
raw GEN: n/a
raw SIM: SimTrack #1128 key=290131

Bremsstrahlung
Vertex 332
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-63.404, -0.495, 267.353, 10.877)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #495 key=430

Annihilation

Vertex 333
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-116.071, -22.707, 202.724, 8.299)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #496 key=431

gamma (22)
Particle 515
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.050, -0.030, 0.091, 0.108)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.000866
direct recHits: 6 missing=0 recoE=0.245166
subgraph simHits: 6 simE=0.000866
subgraph recHits: 6 missing=0 recoE=0.245166
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-49.035, -9.262, 293.500, 0.000)
p4@checkpoint: (0.050, -0.030, 0.091, 0.108)
raw GEN: n/a
raw SIM: SimTrack #1129 key=290137

gamma (22)
Particle 516
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000204
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000204
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-119.311, -31.109, 201.256, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1131 key=290893

gamma (22)
Particle 517
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.095, -0.029, 0.147, 0.178)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 8 simE=0.001753
subgraph recHits: 6 missing=2 recoE=0.115757
direct tracker simHits: 16 dE=0.001697
subgraph tracker simHits: 17 dE=0.001705
raw GEN: n/a
raw SIM: SimTrack #1133 key=290897

PairConversion

Vertex 335
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-52.942, -66.689, 152.592, 6.286)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #499 key=434

Annihilation

Vertex 336
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-53.663, -67.766, 220.776, 8.980)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #500 key=435

e- (11)
Particle 518
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.090, -0.027, 0.140, 0.169)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.001263
direct recHits: 6 missing=1 recoE=0.115757
subgraph simHits: 7 simE=0.001263
subgraph recHits: 6 missing=1 recoE=0.115757
direct tracker simHits: 1 dE=0.000008
subgraph tracker simHits: 1 dE=0.000008
checkpointId: 0
x4@checkpoint: (-59.249, -61.992, 293.500, 0.000)
p4@checkpoint: (0.026, -0.087, 0.133, 0.161)
raw GEN: n/a
raw SIM: SimTrack #1134 key=291096

gamma (22)
Particle 519
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000490
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000490
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-68.035, -102.831, 228.770, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1135 key=291102

e- (11)
Particle 520
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.136, -0.197, 0.388, 0.456)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 14 simE=0.373180
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 15 simE=0.373518
subgraph recHits: 0 missing=15 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-79.624, -94.143, 201.850, 0.000)
p4@checkpoint: (-0.061, -0.231, 0.374, 0.443)
raw GEN: n/a
raw SIM: SimTrack #1137 key=291284

Bremsstrahlung
Vertex 340
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-76.990, -88.043, 191.328, 7.651)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #505 key=440

e+ (-11)
Particle 521
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.175, -0.253, 0.491, 0.579)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 14 simE=0.525076
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 16 simE=0.529287
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-81.523, -92.504, 201.646, 0.000)
p4@checkpoint: (-0.231, -0.183, 0.475, 0.559)
raw GEN: n/a
raw SIM: SimTrack #1138 key=291285

Bremsstrahlung
Vertex 338
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-76.653, -87.515, 190.318, 7.611)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #503 key=438

Bremsstrahlung
Vertex 339
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-77.503, -88.599, 192.596, 7.700)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #504 key=439

gamma (22)
Particle 522
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.002, 0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.004075
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.004075
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-80.685, -93.235, 201.722, 0.000)
p4@checkpoint: (-0.001, -0.002, 0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #1139 key=291286

gamma (22)
Particle 523
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000137
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000137
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-81.112, -92.864, 201.826, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1140 key=291289

gamma (22)
Particle 524
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000338
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000338
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-80.404, -93.477, 201.672, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1141 key=291462

n (2112)
Particle 525
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.026, 0.030, -0.011, 0.940)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000152
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000152
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.730, 78.924, 139.844, 0.000)
p4@checkpoint: (-0.026, 0.030, -0.011, 0.940)
raw GEN: n/a
raw SIM: SimTrack #1142 key=293735

p (2212)
Particle 526
pid: p (2212)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.146, 0.380, -0.136, 1.032)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 4 simE=0.005564
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1143 key=301970

HadronInelastic

Vertex 343
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-67.149, 96.750, 169.548, 9.909)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #508 key=443

n (2112)
Particle 527
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.283, -0.149, 0.265, 1.027)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000066
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000066
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-91.971, 82.124, 192.707, 0.000)
p4@checkpoint: (-0.283, -0.149, 0.265, 1.027)
raw GEN: n/a
raw SIM: SimTrack #1144 key=301971

n (2112)
Particle 528
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.049, 0.036, -0.080, 0.945)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.005143
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.005143
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-71.850, 100.202, 161.924, 0.000)
p4@checkpoint: (-0.049, 0.036, -0.080, 0.945)
raw GEN: n/a
raw SIM: SimTrack #1147 key=301996

gamma (22)
Particle 529
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.001, -0.004, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000421
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000421
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-79.082, 94.599, 158.412, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1148 key=301997

n (2112)
Particle 530
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.015, -0.012, 0.025, 0.940)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-67.657, 103.080, 161.709, 0.000)
p4@checkpoint: (0.002, 0.020, -0.019, 0.940)
raw GEN: n/a
raw SIM: SimTrack #1150 key=301999

gamma (22)
Particle 531
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.003, -0.000, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.004654
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.004654
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-76.002, -97.091, 169.970, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1152 key=302523

gamma (22)
Particle 532
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.002, -0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.010467
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.010467
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-102.452, 68.603, 163.315, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1153 key=302524

gamma (22)
Particle 533
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002269
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002269
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (118.899, 32.647, 246.410, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1155 key=302529

gamma (22)
Particle 534
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001173
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001173
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-81.748, 92.305, 22.195, 0.000)
p4@checkpoint: (-0.001, 0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1162 key=322983

gamma (22)
Particle 535
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.001, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000845
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000845
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-24.481, -120.845, 60.210, 0.000)
p4@checkpoint: (-0.000, -0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1163 key=322985

gamma (22)
Particle 536
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.002, 0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000120
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000120
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-77.403, 95.977, 54.583, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1164 key=337688

gamma (22)
Particle 537
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000795
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000795
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.295, 1.074, 222.605, 0.000)
p4@checkpoint: (0.001, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1167 key=337700

n (2112)
Particle 538
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.010, 0.041, -0.066, 0.943)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000066
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000066
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.980, 70.752, -20.115, 0.000)
p4@checkpoint: (-0.010, 0.041, -0.066, 0.943)
raw GEN: n/a
raw SIM: SimTrack #1168 key=337753

gamma (22)
Particle 539
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001010
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001010
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.720, 75.191, 13.567, 0.000)
p4@checkpoint: (0.000, 0.001, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1172 key=337766

n (2112)
Particle 540
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.258, 0.338, -0.013, 1.031)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=0.014870
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=0.014870
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (107.020, 61.234, -47.373, 0.000)
p4@checkpoint: (0.258, 0.338, -0.013, 1.031)
raw GEN: n/a
raw SIM: SimTrack #1174 key=339768

gamma (22)
Particle 541
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000449
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000449
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (101.355, -70.214, 71.318, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1175 key=339770

gamma (22)
Particle 542
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000105
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000105
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.286, -1.879, 46.740, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1176 key=339771

connector

Particle 543
genEvent: 0
hasGen: no hasSim: no
isRoot: no isLeaf: no
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 1
nParents: 0 nChildren: 10
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 4710 simE=426.584503
subgraph recHits: 959 missing=3751 recoE=71.480919
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 741 dE=0.235754
raw GEN: n/a
raw SIM: n/a

ISR/upstream
Vertex 353
ISR/upstream (genEvent=0, eid=0)
domain: Internal
genEvent: 0
hasGen: no hasSim: no
isSource: no isSink: no
x4: (0.000, 0.000, 3.451, 0.198)
nIn: 1 nOut: 10
raw GEN: n/a
raw SIM: n/a

interaction

Vertex 352
interaction (genEvent=0, eid=0)
domain: Internal
genEvent: 0
hasGen: no hasSim: no
isSource: yes isSink: no
x4: (0.000, 0.000, 3.451, 0.198)
nIn: 0 nOut: 1
raw GEN: n/a
raw SIM: n/a